

Unit 4

FORCES, DENSITY AND PRESSURE

59 items: 26 from official papers (113 marks), 33 from the booklet

Syllabus sections in this unit

- 4.1 Turning effects of forces
- 4.2 Equilibrium of forces
- 4.3 Density and pressure

Unit 4: Forces, density and pressure

1. M/J 26/P22/Q2/a,b

- 2 (a) State the conditions required for an object to be in equilibrium.

[2]

- (b) A block rests in equilibrium on the centre of a uniform horizontal plank. The block is attached to two ropes, each of which passes over a fixed pulley and then attaches to an end of the plank as shown in Figure 2.1.

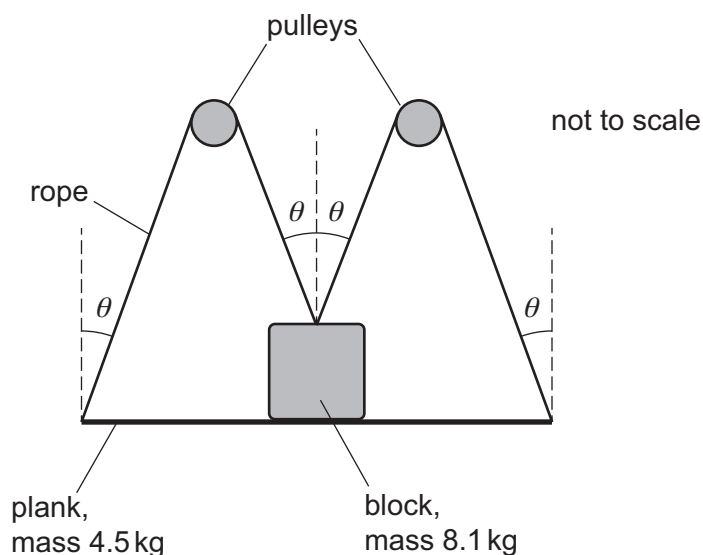


Figure 2.1

Where the ropes attach to the block and to the plank, the angle between the ropes and the vertical is θ . There is negligible friction between the pulleys and the ropes.

The mass of the block is 8.1 kg. The mass of the plank is 4.5 kg.

Figure 2.2 shows the directions of the forces acting on the plank and the block.

The weight of the plank is W_P , the weight of the block is W_B , the tension in each rope is T and the forces that the plank and the block exert on each other each have magnitude R .

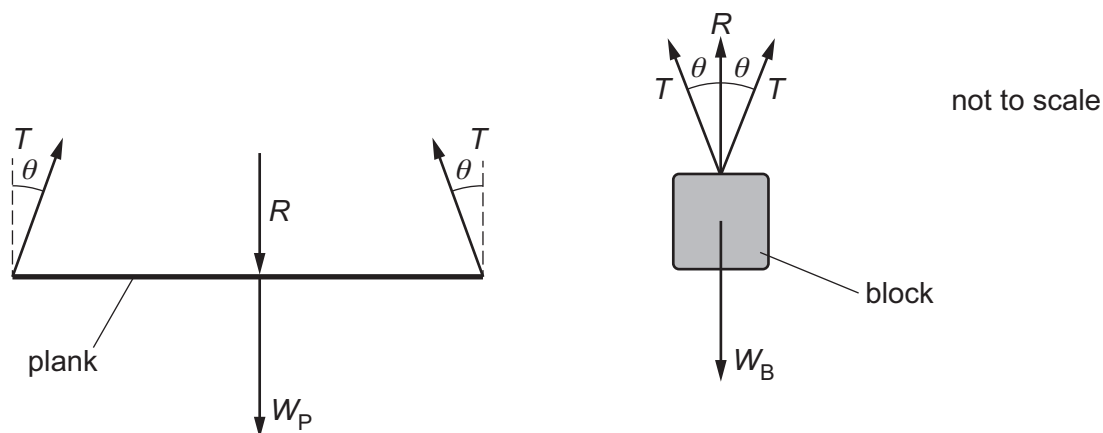


Figure 2.2

- (i) Use Figure 2.2 to show that

$$R + W_P = W_B - R.$$

[2]

- (ii) Use the equation in 2(b)(i) to calculate the magnitude of R .

$$R = \text{N [2]}$$

2. M/J 26/P23/Q2/b(i)

- (b) A long, thin thread of high-density polythene has a uniform cross-section. The thread is suspended from one end so that it hangs vertically.

The thread has a mass of 0.018 kg and a cross-sectional area of $2.6 \times 10^{-6} \text{ m}^2$.
High-density polythene has a density of 960 kg m^{-3} and a Young modulus of $8.0 \times 10^8 \text{ Pa}$.

- (i) Show that the length of the thread is 7.2 m .

[2]

3. M/J 26/P24/Q4/c

- 4 An object is held at rest just below the surface of a liquid of uniform density as shown in Figure 4.1.

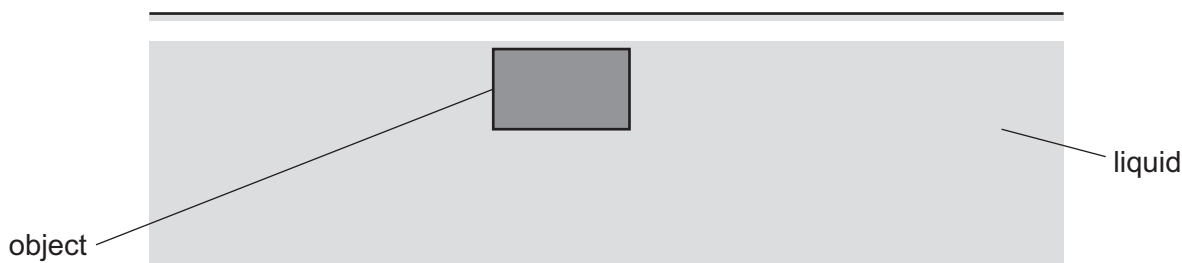


Figure 4.1

The object is released and accelerates downwards until it reaches terminal velocity.

The magnitude of the weight of the object is W .

The magnitude of the upthrust acting on the object is U .

The magnitude of the drag force acting on the object is D .

- (c) The object has a mass of 0.070 kg and a volume of $1.0 \times 10^{-5} \text{ m}^3$.

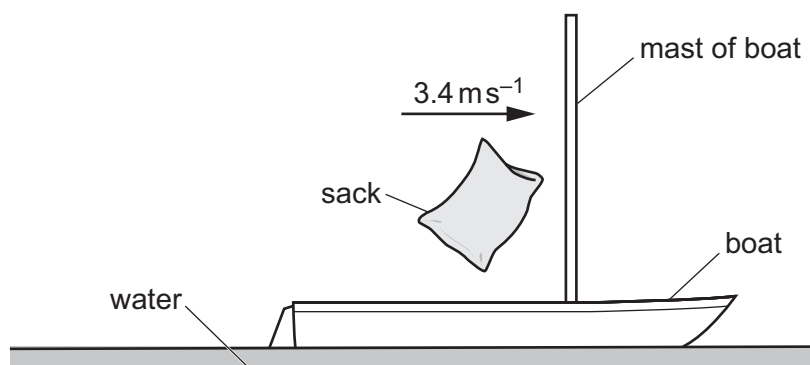
The density of the liquid is 1100 kg m^{-3} .

Calculate D when the object is at terminal velocity.

$$D = \text{N [3]}$$

4. MAR 26/P22/Q2/c

- (b) A stationary boat of mass 160 kg floats on water. The mast of the boat is vertical. A sack of mass 23 kg is thrown at the boat, as shown in Figure 2.1.

**Figure 2.1**

Immediately before colliding with the mast of the boat, the sack is travelling horizontally with a speed of 3.4 ms^{-1} .

The sack collides with the mast and remains in contact with it, without rebounding.

Assume that resistive forces acting on the boat during the collision are negligible.

- (i) Calculate the horizontal speed v of the sack and boat immediately after the collision.

$$v = \quad \text{ms}^{-1} \quad [2]$$

- (ii) The total kinetic energy of the sack and boat immediately before the collision is E_1 . The total kinetic energy of the sack and boat immediately after the collision is E_2 .

Determine the ratio $\frac{E_2}{E_1}$.

$$\text{ratio} = \quad [3]$$

- (iii) State and explain whether the collision between the sack and boat is elastic or inelastic.

[1]

- (c) After a short time, the boat in **2(b)** is at rest again.

- (i) Show that the upthrust acting on the boat with the sack is 1800 N.

[1]

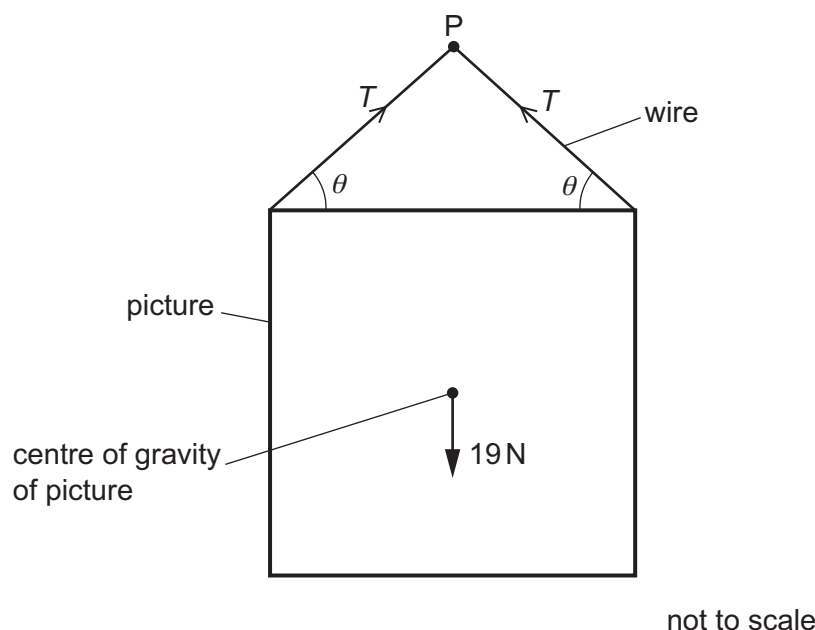
- (ii) The boat displaces water with a volume of $1.80 \times 10^5 \text{ cm}^3$.

Calculate the density of the water. Give your answer to 3 significant figures.

$$\text{density} = \quad \text{kg m}^{-3} \quad [2]$$

5. MAR 26/P22/Q4/a(i,ii,iii)

- 4 A uniform square picture is attached to a wire and hangs in equilibrium from a pin at point P, as shown in Figure 4.1.

**Figure 4.1**

The picture has weight 19 N and its centre of gravity is directly below P.

The top side of the picture is horizontal. The wire forms an angle θ with the horizontal. The tension T in the wire is 13 N.

- (a) (i) Define the moment of a force about a point.

[1]

- (ii) Explain why the moment of the weight about P is zero.

[1]

- (iii) Calculate θ .

$\theta =$ ° [2]

6. O/N 25/P21/Q2/a,c(i)

- 2 (a) (i) Define pressure.

[1]

- (ii) Explain how hydrostatic pressure results in an upthrust force acting on a solid object immersed in a liquid.

[2]

- (c) The oil in (b) has a density of 920 kg m^{-3} and a viscosity of 4.7 in SI units.

The steel ball has a mass of $2.4 \times 10^{-3} \text{ kg}$ and a radius of $4.2 \times 10^{-3} \text{ m}$.

- (i) Show that the upthrust force acting on the ball is $2.8 \times 10^{-3} \text{ N}$.

[1]

7. O/N 25/P22/Q1/b(i,ii)

1 Scientists are investigating the variation in air pressure at different locations on a mountain.

- (b) (i) At one location, the density of the air is 1.1 kg m^{-3} . A spherical weather balloon is filled with a gas and released from rest. The balloon has radius 0.90 m .

Calculate the upthrust acting on the balloon when it is released.

upthrust = N [2]

- (ii) Explain why an upthrust acts on the balloon.

[2]

8. O/N 25/P22/Q2/a

2 A spacecraft in deep space uses jets of hot gas from its thrusters to change its velocity. Fig. 2.1 shows a side view of the spacecraft and some of its thrusters.

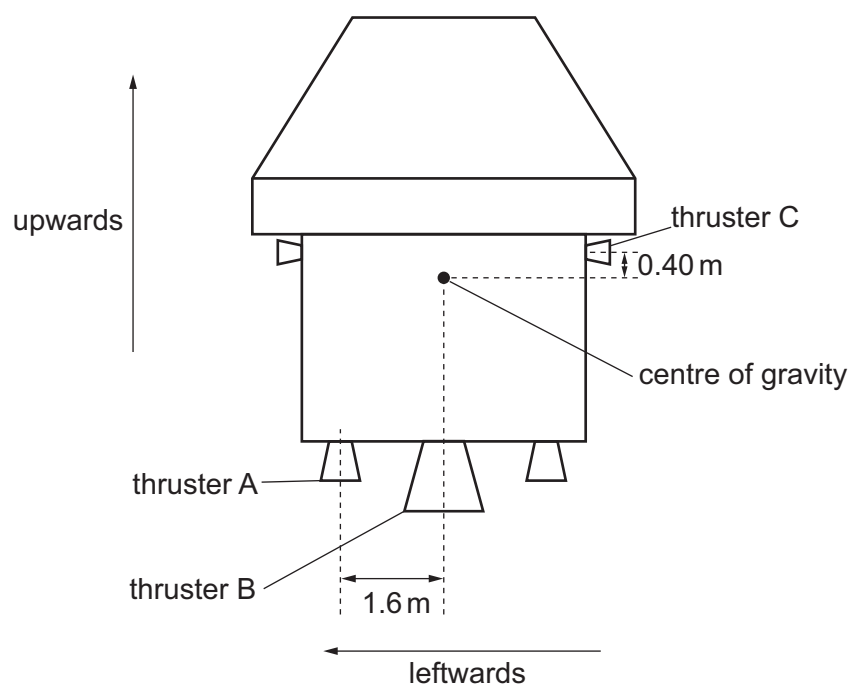


Fig. 2.1 (not to scale)

Thruster A is a distance of 1.6 m leftwards from the centre of gravity of the spacecraft. Thruster C is a distance of 0.40 m upwards from the centre of gravity of the spacecraft.

Thrusters A and B can produce forces on the spacecraft in the upwards direction only.

Thruster C can produce a force on the spacecraft in the leftwards direction only.

All the thrusters shown produce forces entirely in the same plane as the centre of gravity.

- (a) (i) Thruster A is activated, producing a force of 60 N upwards on the spacecraft. Thruster C is also activated, producing a force of 220 N in the leftwards direction on the spacecraft.

Calculate the resultant moment due to these forces about the centre of gravity.

resultant moment = Nm [2]

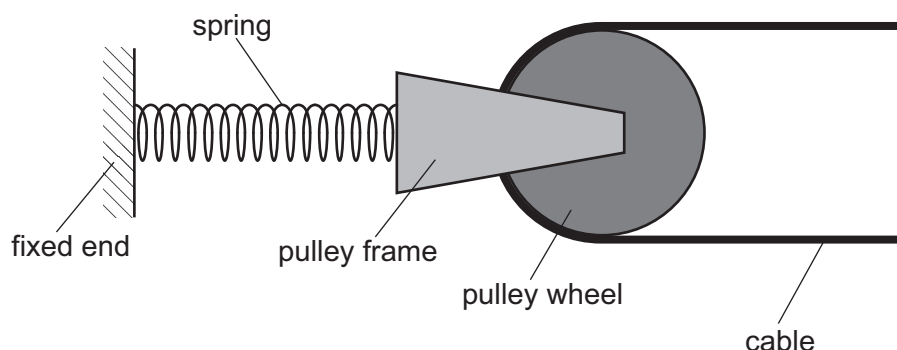
- (ii) State and explain whether the forces from A and C are a couple.

[1]

9. O/N 25/P22/Q3/a,b(i)

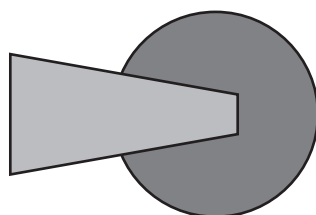
- 3 A spring is fixed at one end and attached to the frame of a pulley at the other end. A cable is passed around the wheel of the pulley. The spring is stretched to a fixed length using the cable and pulley.

Fig. 3.1 shows the view from above of the spring, cable and pulley.

**Fig. 3.1**

The spring obeys Hooke's law and has a spring constant k of 250 N m^{-1} . A force F acts on the spring. The tension in the cable is T . The pulley is in equilibrium.

- (a) On Fig. 3.2, draw labelled arrows to show the directions of the forces acting on the pulley.

**Fig. 3.2**

[2]

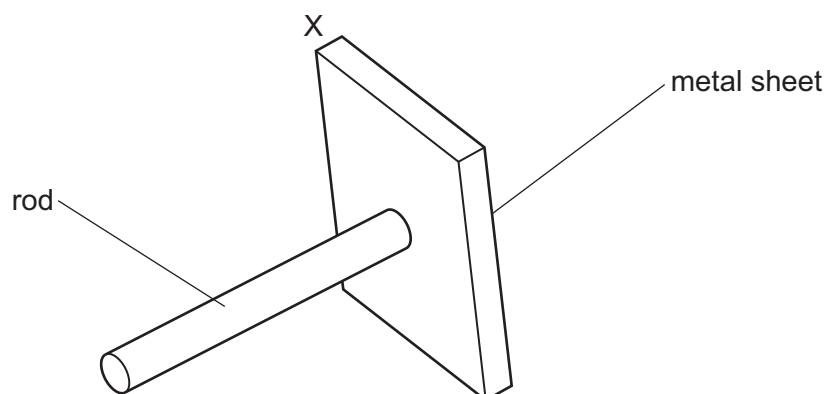
- (b) The force F is 110 N .

- (i) Determine T .

$T =$ N [1]

10. O/N 25/P24/Q2/a,b,c,d,e

- 2 Fig. 2.1 shows a square metal sheet of non-uniform density, with a thin wooden rod fixed at its centre. One of the corners of the sheet is labelled X.

**Fig. 2.1**

The rod has negligible mass. The mass of the metal sheet is 2.8 kg.
The rod is supported so that the rod is horizontal and the metal sheet is vertical.

- (a) Define the torque of a couple.

[2]

- (b) When the rod is supported in such a way that it can rotate freely within its support, the sheet hangs in equilibrium with point X vertically above the rod, as shown in Fig. 2.2.

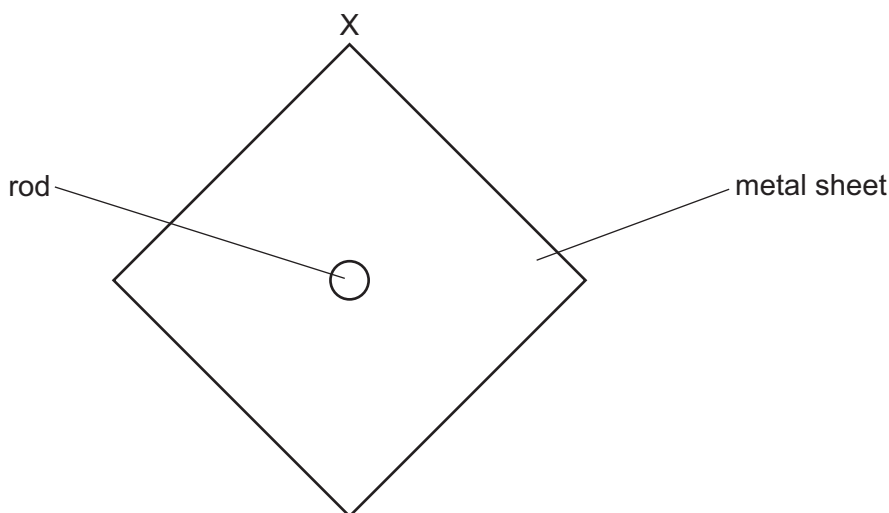


Fig. 2.2

On Fig. 2.2, draw a line to indicate the range of possible positions for the centre of gravity of the metal sheet.

[1]

- (c) When a torque of 3.3 N m is applied to the rod, the sheet is held in equilibrium with two of its edges horizontal, as shown in Fig. 2.3. Point X is at the top-left corner.

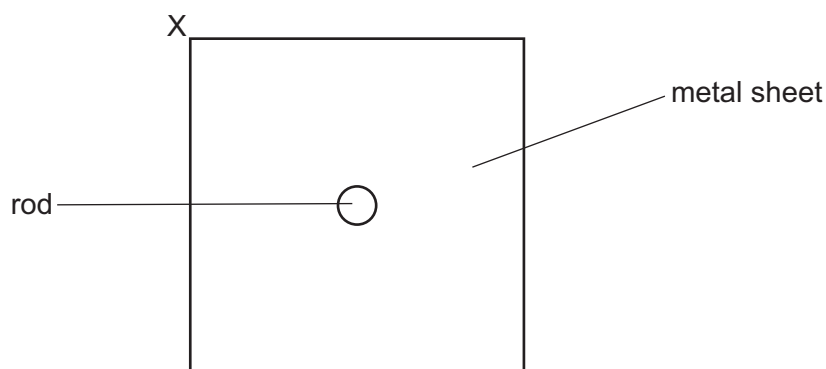


Fig. 2.3

- (i) Explain whether the torque applied to the rod to hold the sheet in equilibrium is clockwise or anticlockwise.

[1]

- (ii) Show that the centre of gravity of the sheet has a horizontal displacement of 0.12 m from the rod.

[1]

- (d) The square metal sheet has an average density of 3000 kg m^{-3} and a uniform thickness of 4.0 mm.

Show that the side length of the sheet is 0.48 m.

[3]

- (e) Use the answer in (b) and the information in (c) and (d) to determine the position of the centre of gravity of the sheet. Indicate this position on Fig. 2.3 with a point labelled Y. [2]

11. M/J 25/P21/Q2/a,b

- 2 (a) State the principle of moments.

[2]

- (b) Three objects A, B and C are placed on a horizontal beam. The beam is in equilibrium, as shown in Fig. 2.1.

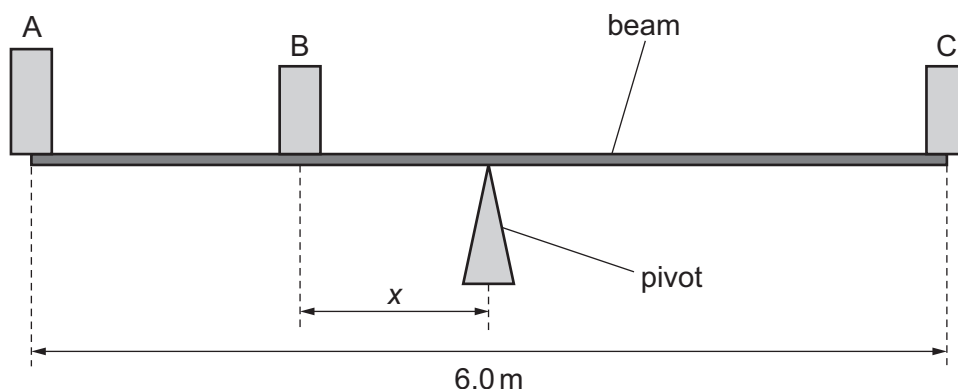


Fig. 2.1 (not to scale)

The beam is uniform and has length 6.0 m.

The pivot is at the midpoint of the beam.

Object A has mass 60 kg and is at one end of the beam.

Object B has mass 45 kg and is at a distance x from the pivot.

Object C has mass 80 kg and is at the other end of the beam.

Calculate x .

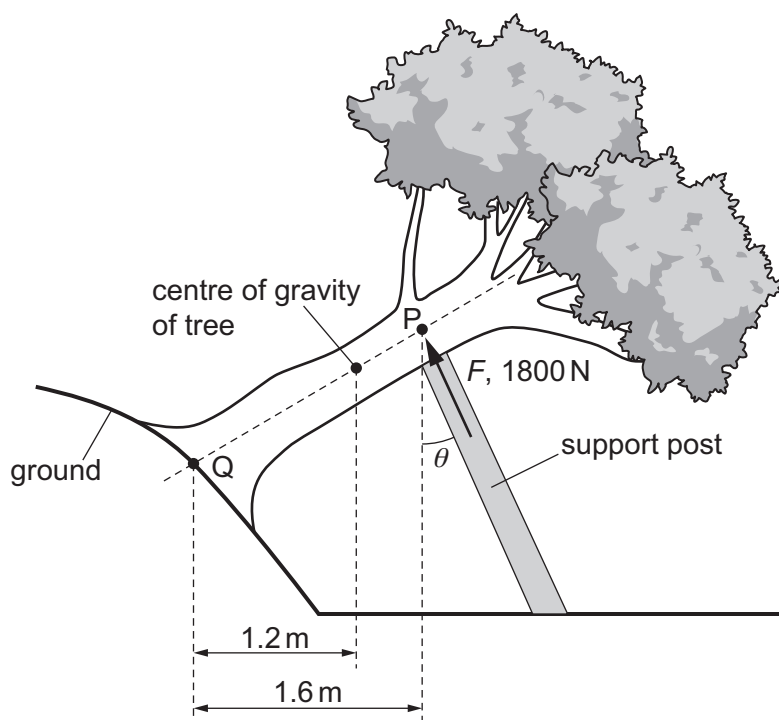
$x =$ m [3]

12. M/J 25/P22/Q2/a,b

- 2 (a) Define the moment of a force about a point.

[1]

- (b) A tree of mass 270 kg grows out of sloping ground and is supported by a post, as shown in Fig. 2.1.

**Fig. 2.1** (not to scale)

The ground applies a total force R on the tree at point Q.

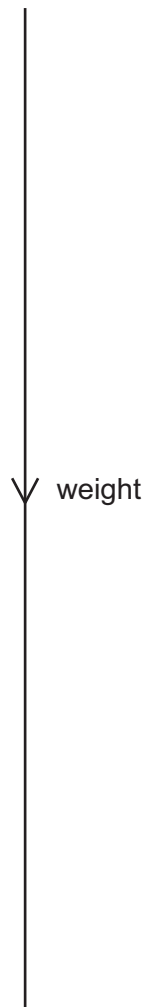
The centre of gravity of the tree is a horizontal distance of 1.2 m from Q.

The post applies a force F of 1800 N perpendicular to the line PQ. The line of action of F passes through point P at an angle θ to the vertical. P is a horizontal distance of 1.6 m from Q. The tree is in equilibrium and all forces act on the tree in the same plane.

- (i) By taking moments about point Q, show that θ is 25° .

[3]

- (ii) On Fig. 2.2, draw a labelled scale vector triangle to represent the forces acting on the tree. The weight of the tree has been drawn to scale.

**Fig. 2.2**

[2]

- (iii) The tree exerts a pressure of 150 kPa on the top of the post.

Determine the surface area of the tree in contact with the post.

area =

m² [2]

13. M/J 25/P23/Q2/a,b(i)

- 2 (a)** Define the moment of a force about a pivot.

[1]

- (b)** Three objects A, B and C are placed on a horizontal beam. The beam is in equilibrium, as shown in Fig. 2.1.

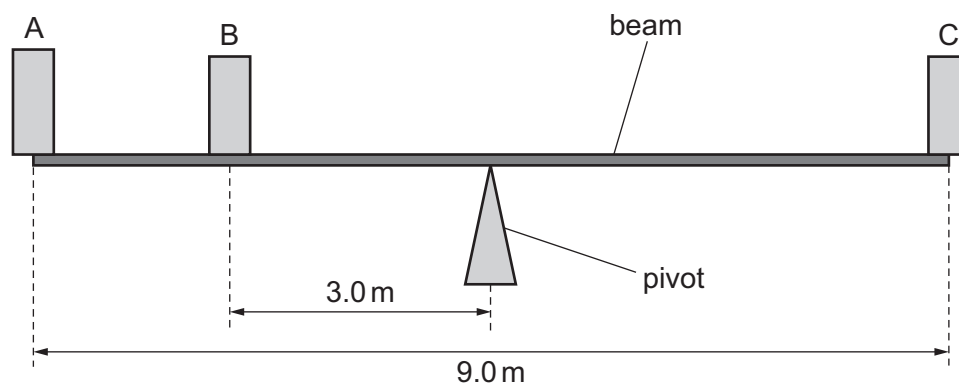


Fig. 2.1 (not to scale)

The beam is uniform and has length 9.0 m.

A pivot is at the midpoint of the beam.

Object A has mass 90 kg and is at one end of the beam.

Object B has mass m and is a distance of 3.0 m from the pivot.

Object C has mass 150 kg and is at the other end of the beam.

- (i)** Calculate m .

$m =$

kg [3]

14. M/J 25/P24/Q1/a,b(i,iii)

- 1 (a) Define the moment of a force.

[1]

- (b) A trapdoor has a hinge at end A, as shown in Fig. 1.1.

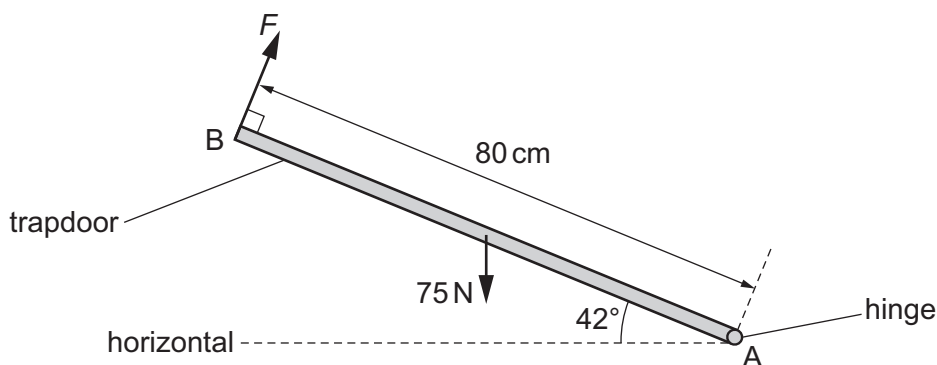


Fig. 1.1 (not to scale)

The trapdoor has length 80 cm and weight 75 N. The mass of the trapdoor is uniformly distributed along its length.

A force F acts at right angles to the trapdoor at end B so that the trapdoor is held in equilibrium at an angle of 42° to the horizontal.

- (i) State the principle of moments.

[2]

- (iii) Calculate the magnitude of the force F .

$F =$

N [2]

15. MAR 25/P22/Q1/b(i)

- (b) Two solid cubes, A and B, are measured to determine the density of their materials.

Table 1.1 shows the measurements for cube A.

Table 1.1

quantity	measurement
length of side	(1.53 ± 0.01) cm
mass	(31.3 ± 0.5) g

- (i) Show that the calculated density of the material of cube A is $8.7 \times 10^3 \text{ kg m}^{-3}$.

[2]

16. MAR 25/P22/Q2/a,b

- 2 (a) State the principle of moments.

[2]

- (b) A solid plastic cylinder floats in water. It is used to support one end of a horizontal uniform beam AB as shown in Fig. 2.1.

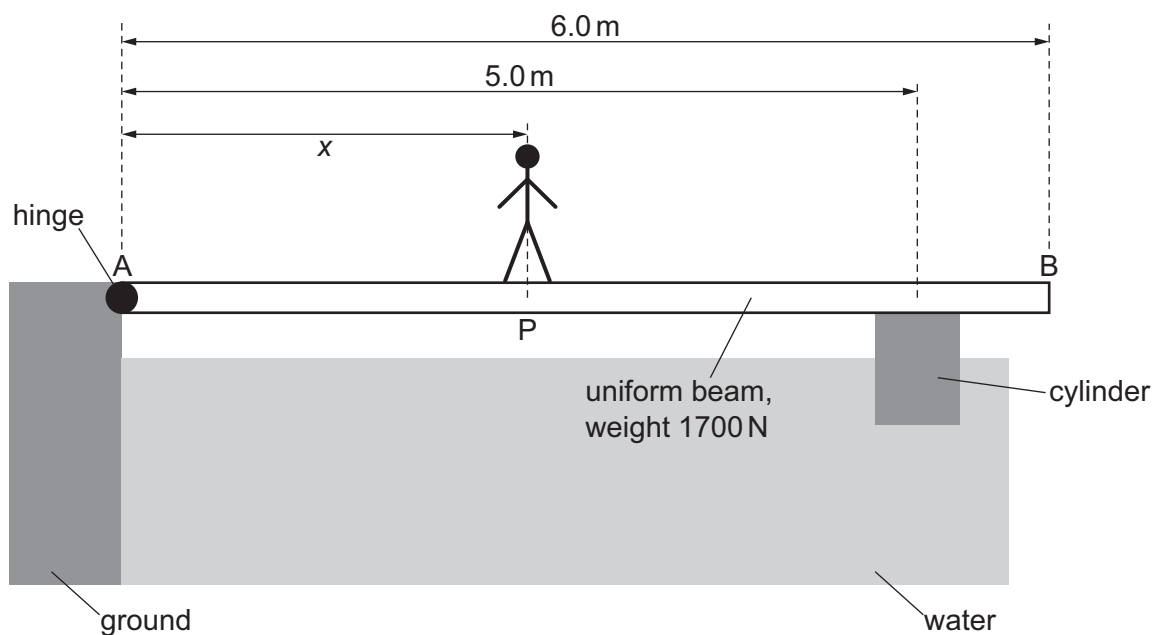


Fig. 2.1 (not to scale)

The beam has length 6.0 m and weight 1700 N. The beam is attached to solid ground with a hinge at end A.

The cylinder is floating vertically in the water. The top of the cylinder is attached at its centre to the beam at a horizontal distance of 5.0 m from end A. The cylinder applies a vertical force of 1300 N to the beam.

A person of weight 660 N stands on the beam at point P.

The beam AB is in equilibrium.

- (i) By taking moments about end A, determine the distance x from A to P.

distance = _____ m [3]

- (ii) The bottom of the cylinder is submerged in the water to depth y as shown in Fig. 2.2. The beam is still attached to the cylinder but not shown.

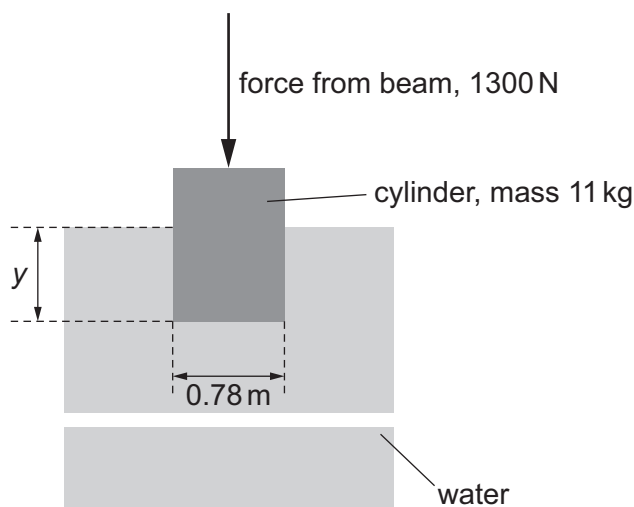


Fig. 2.2 (not to scale)

The cylinder has mass 11 kg and diameter 0.78 m. The beam exerts a vertical force of 1300 N on the cylinder. The cylinder is in equilibrium.

Show that the upthrust acting on the cylinder is 1400 N.

[1]

- (iii) The water has density 990 kg m^{-3} .

Calculate the depth y .

$y =$ m [2]

- (iv) The person can stand anywhere between A and B.

On Fig. 2.3, sketch the variation of the depth of the bottom of the cylinder with the distance of the person from A, for distances between 0 and 6.0 m. Numerical values are not required.

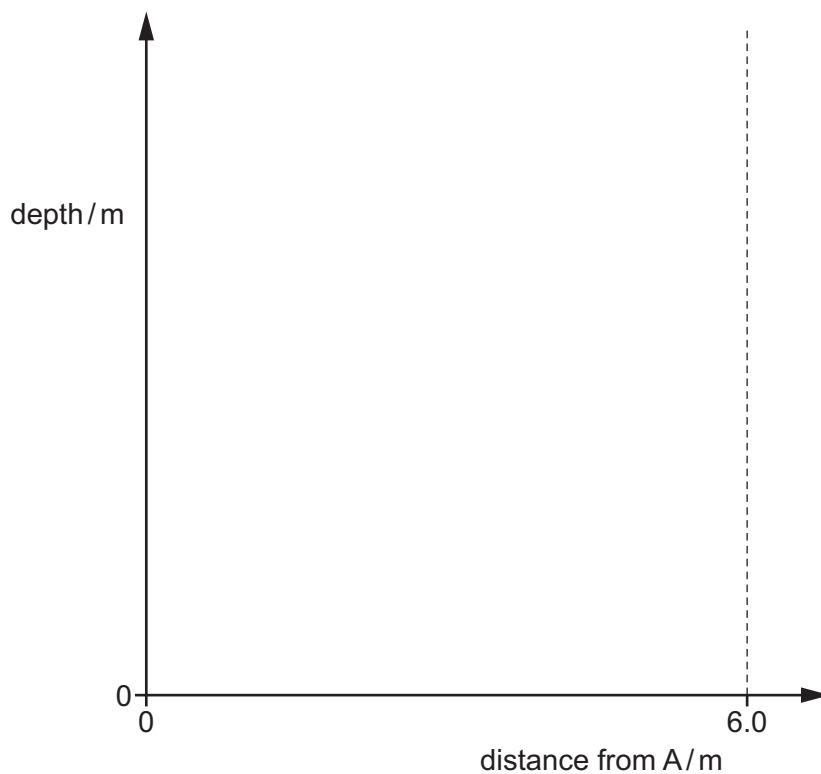
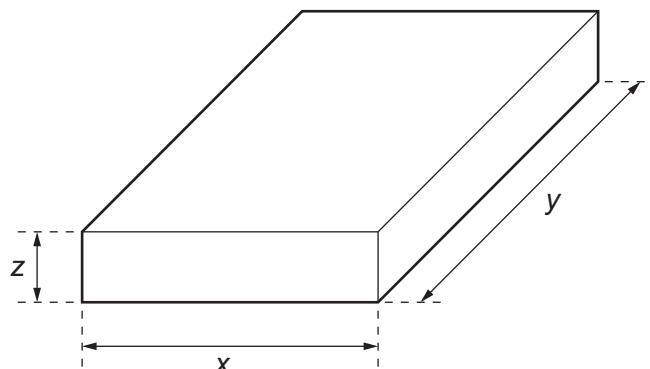


Fig. 2.3

[2]

17. O/N 24/P21/Q1/a,b(i)**1 (a)** Define density.

[1]

(b) Fig. 1.1 shows a cuboidal glass block.**Fig. 1.1** (not to scale)

A student measures the mass m of the block and the side lengths x , y and z . The measurements are shown in Table 1.1.

Table 1.1

quantity	measurement
m	$(0.243 \pm 0.001) \text{ kg}$
x	$(5.41 \pm 0.01) \text{ cm}$
y	$(11.09 \pm 0.01) \text{ cm}$
z	$(1.62 \pm 0.01) \text{ cm}$

(i) Determine the density of the glass.

density =

 kg m^{-3} [2]

18. O/N 24/P22/Q3/b,c(i)

(b) State the conditions necessary for an object to be in equilibrium.

[2]

(c) A wire is used to hold a uniform shelf AB horizontally in equilibrium as shown in Fig. 3.2.

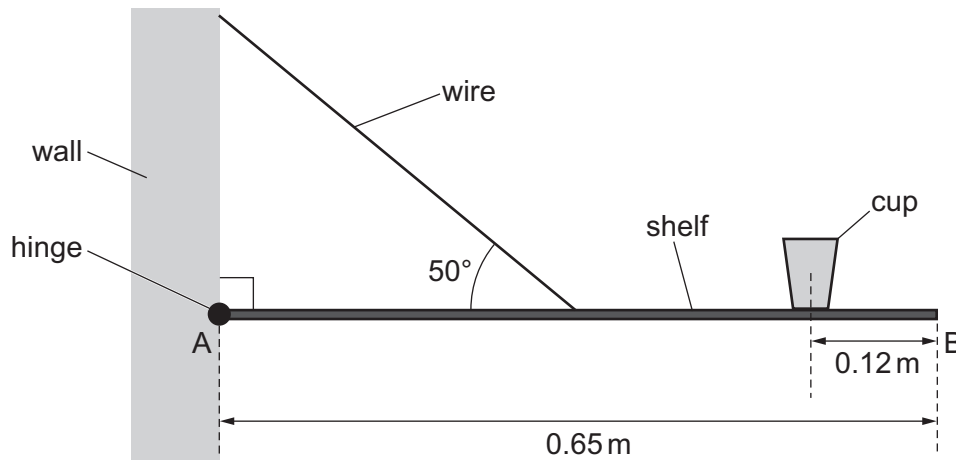


Fig. 3.2 (not to scale)

The wire is connected to the midpoint of shelf AB at an angle of 50° to the horizontal. The shelf is attached to a wall by a hinge at A. The length of shelf AB is 0.65 m and its weight is 33 N.

A cup of weight 1.5 N rests on the shelf with its centre of gravity at a horizontal distance of 0.12 m from B.

(i) By taking moments about A, determine the tension in the wire.

tension =

N [3]

19. O/N 24/P23/Q3/a,b,c

3 (a) State the principle of moments.

[1]

(b) A rigid uniform beam rests on a pivot at its centre, as shown in Fig. 3.1.

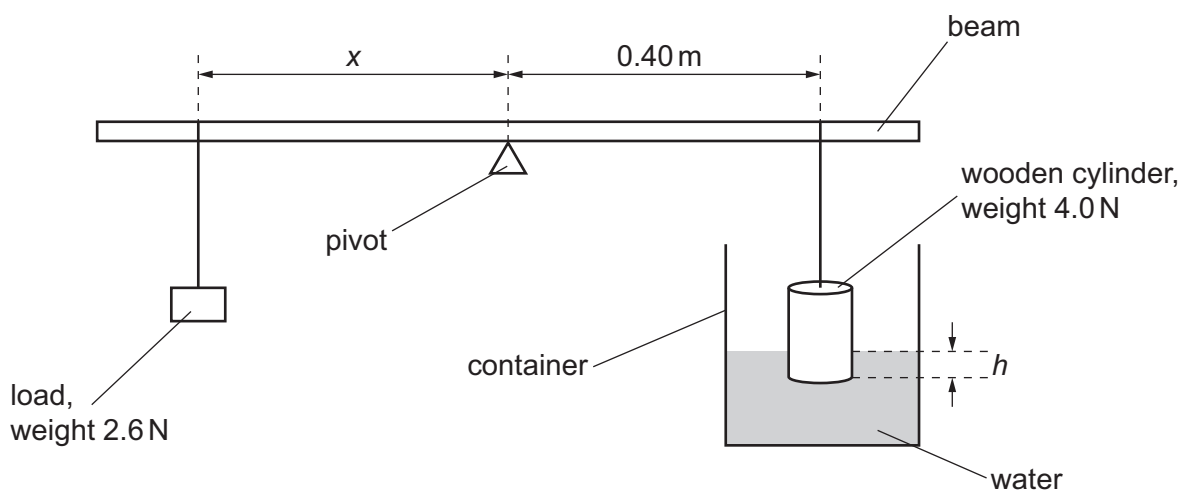


Fig. 3.1 (not to scale)

A load of weight 2.6 N is suspended from the beam at distance x from the pivot.

A wooden cylinder of weight 4.0 N is suspended from the beam at a distance of 0.40 m from the pivot on the opposite side of the pivot to the load. The cylinder rests in a container of water. The lower part of the cylinder is immersed in the water to depth h .

Initially, h is equal to 0.10 m and x is equal to 0.40 m. The system is in equilibrium.

- (i) Use the principle of moments to show that the upthrust U exerted by the water on the cylinder is 1.4 N.

[2]

- (ii) The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Calculate the area A of the circular cross-section of the cylinder.

$$A = \quad \text{m}^2 \quad [3]$$

- (c) More water is gradually added to the container in (b), so that depth h in Fig. 3.1 gradually increases. The length x is continuously adjusted so that the system remains in equilibrium.

On Fig. 3.2, sketch the variation of x with h . Use the space below for any working.

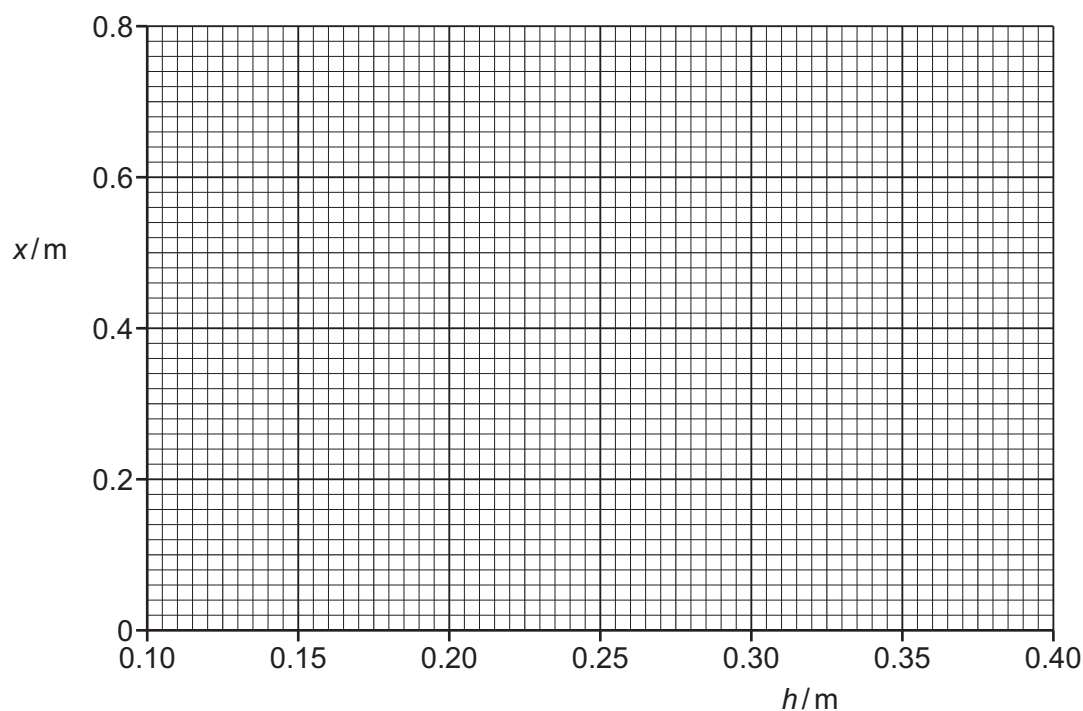
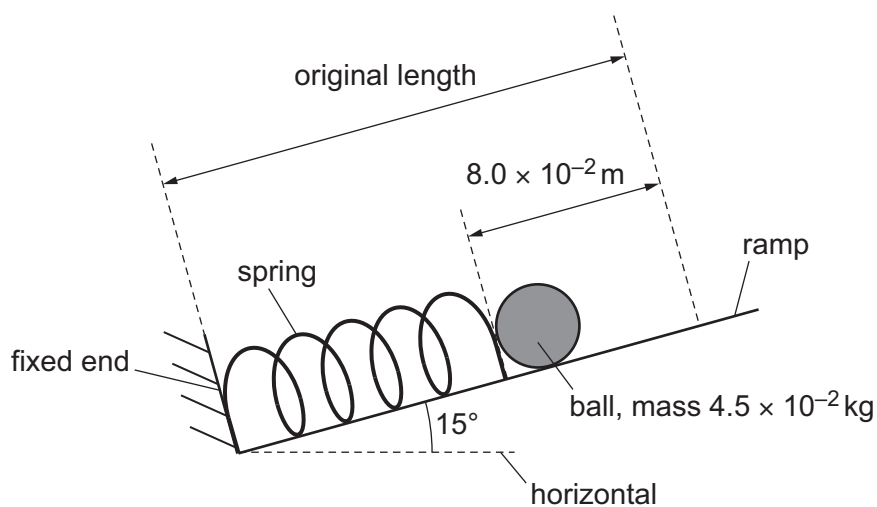


Fig. 3.2

[3]

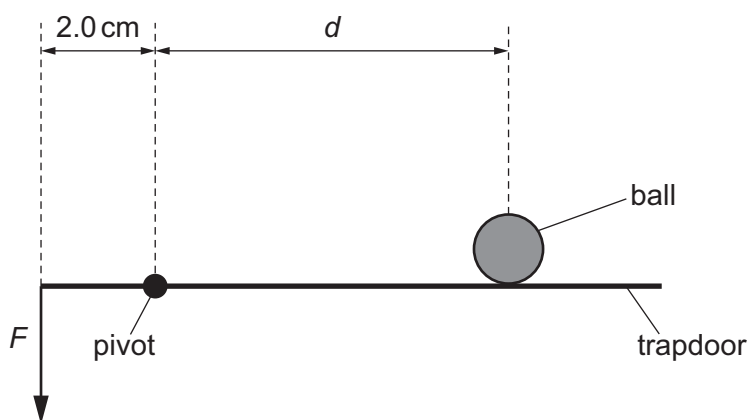
20. M/J 24/P22/Q4/c

- 4 A pinball machine uses a spring to launch a small metal ball of mass $4.5 \times 10^{-2} \text{ kg}$ up a ramp. The spring is compressed by $8.0 \times 10^{-2} \text{ m}$ and held in equilibrium, as shown in Fig. 4.1.

**Fig. 4.1** (not to scale)

The ramp is at an angle of 15° to the horizontal.

- (c) The ball comes to rest on a horizontal trapdoor of negligible mass at a distance d from its pivot.
A force F acts vertically downwards at a distance of 2.0 cm from the pivot, as shown in Fig. 4.2.

**Fig. 4.2** (not to scale)

- (i) The trapdoor is in equilibrium when F is 1.7 N .

Calculate d .

$d =$ m [2]

- (ii) Force F is decreased from 1.7 N .

State the direction of the resultant moment about the pivot on the trapdoor.

[1]

21. M/J 24/P23/Q1/d(i)

- 1** The drag force F_D acting on a sphere falling through a liquid is given by

$$F_D = 6\pi\eta r v$$

where r is the radius of the sphere,
 v is the speed of the sphere in the liquid and
 η is a property of the liquid called the viscosity.

- (d) (i)** The density of the liquid is 920 kg m^{-3} .

Show that the upthrust acting on the sphere is 1.0 N .

[2]

22. MAR 24/P22/Q2/c(ii)

(b) An Olympic diver stands on a platform above a pool of water, as shown in Fig. 2.1.

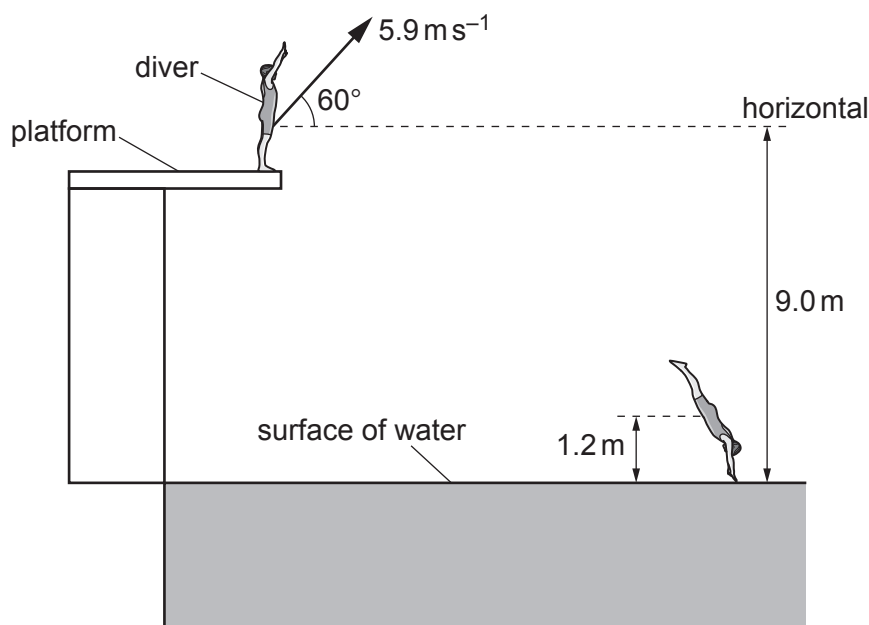


Fig. 2.1 (not to scale)

When the diver is on the platform his centre of gravity is a vertical height of 9.0 m above the surface of the water. The diver jumps from the platform with a velocity of 5.9 m s^{-1} at an angle of 60° to the horizontal.

Air resistance is negligible.

When the diver hits the surface of the water, his centre of gravity is a vertical height of 1.2 m above the surface of the water.

Calculate the speed of the diver at the instant he hits the surface of the water.

speed = m s^{-1} [3]

(c) The diver in (b) enters the water and decelerates.

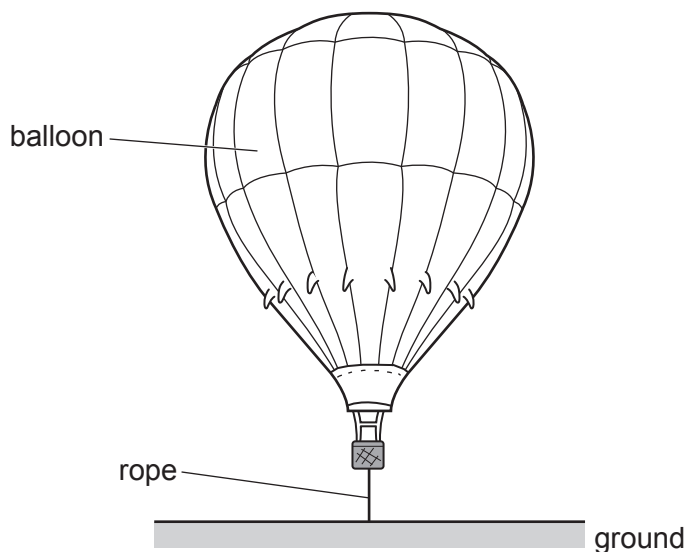
(ii) The diver has a volume of $7.5 \times 10^{-2} \text{ m}^3$. The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Show that the upthrust acting on the diver when he is entirely underwater is 740 N.

[1]

23. O/N 23/P21/Q2/a(i,ii)

- 2 A hot-air balloon floats just above the ground. The balloon is stationary and is held in place by a vertical rope, as shown in Fig. 2.1.

**Fig. 2.1**

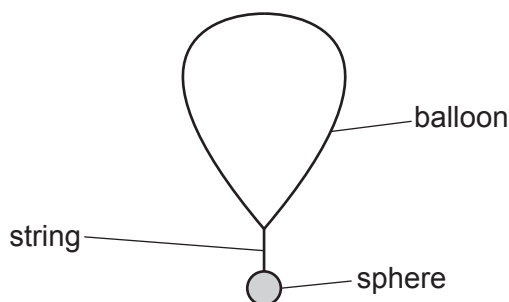
The balloon has a weight W of $3.39 \times 10^4 \text{ N}$. The tension T in the rope is $4.00 \times 10^2 \text{ N}$.
Upthrust U acts on the balloon.
The density of the surrounding air is 1.23 kg m^{-3} .

- (a) (i) On Fig. 2.1, draw labelled arrows to show the directions of the three forces acting on the balloon. [2]
- (ii) Calculate the volume, to three significant figures, of the balloon.

volume = m^3 [3]

24. O/N 23/P22/Q2/a

- 2 A high-altitude balloon is stationary in still air. A solid sphere is suspended from the balloon by a string, as shown in Fig. 2.1.

**Fig. 2.1 (not to scale)**

The volume of the balloon is 7.5 m^3 . The total weight of the balloon, string and sphere is 65 N . The upthrust acting on the string and sphere is negligible.

- (a) Calculate the density of the air surrounding the balloon.

density = kg m^{-3} [2]

25. O/N 23/P23/Q1/b(ii)

- (b) A uniform cylinder has diameter D , length L and mass M .
The density ρ of the cylinder is given by

$$\rho = \frac{4M}{\pi D^2 L}.$$

Table 1.2 shows the data obtained from an experiment to determine the density of the cylinder.

Table 1.2

quantity	measurement	percentage uncertainty
D	$(26.2 \pm 0.1)\text{mm}$	%
L	$(162 \pm 1)\text{mm}$	%
M	$(247 \pm 1)\text{g}$	0.4%

- (ii) Calculate the density of the cylinder. Give your answer to three significant figures.

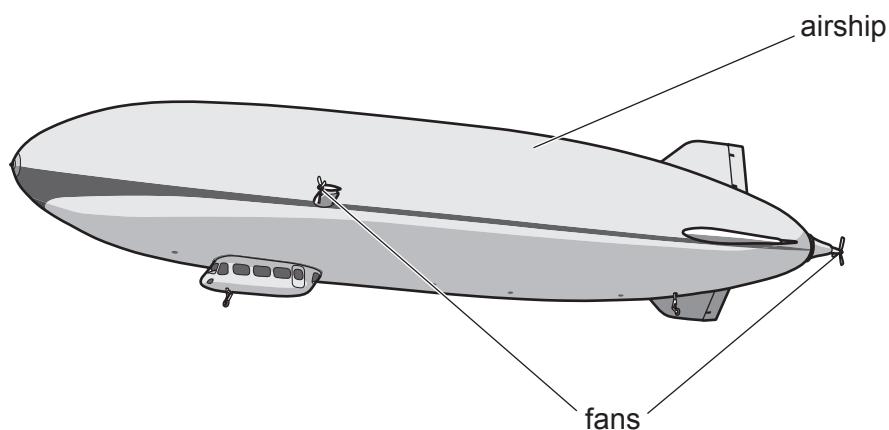
density = kg m^{-3} [2]

26. O/N 23/P23/Q3/a,b

- 3 (a) State the conditions for a system to be in equilibrium.

[2]

- (b) Fig. 3.1 shows an airship in flight. The airship is propelled by identical fans that can be angled to control the motion of the airship.

**Fig. 3.1**

The upthrust on the airship is 93 000 N.
The density of the surrounding air is 1.2 kg m^{-3} .

- (i) Calculate the volume of air displaced by the airship.

volume = m^3 [1]

- (ii) When fully loaded, the weight of the airship is greater than the upthrust.
To maintain horizontal flight, the fans provide a total vertical force of $3.0 \times 10^3 \text{ N}$ upwards on the airship.

Calculate the mass of the airship.

mass = kg [2]

27. M/J 23/P22/Q2

booklet (scan + OCR)

- (a) State what is meant by the centre of gravity of an object.

[1]

- (b) Two blocks are on a horizontal beam that is pivoted at its centre of gravity, as shown in Fig. 2.1.

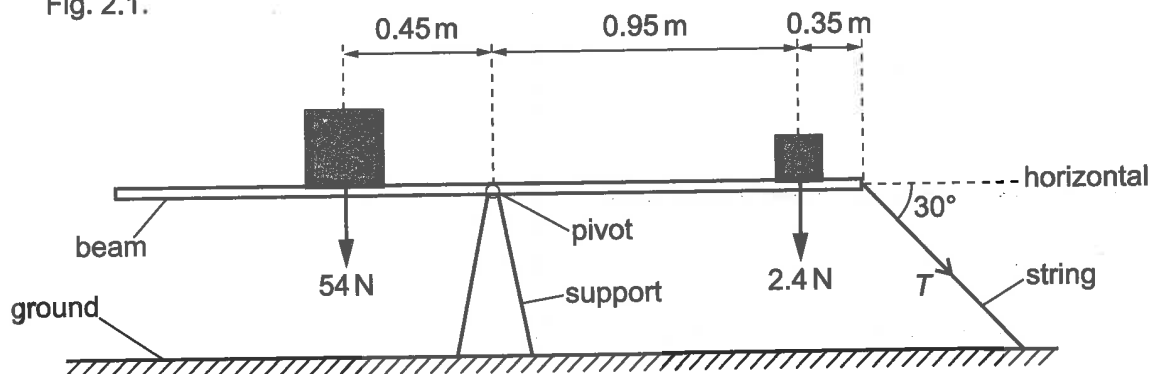


Fig. 2.1 (not to scale)

A large block of weight 54 N is a distance of 0.45 m from the pivot. A small block of weight 2.4 N is a distance of 0.95 m from the pivot and a distance of 0.35 m from the right-hand end of the beam.

The right-hand end of the beam is connected to the ground by a string that is at an angle of 30° to the horizontal. The beam is in equilibrium.

- (i) By taking moments about the pivot, calculate the tension T in the string.

$$T = \dots\dots\dots \text{ N [3]}$$

- (ii) The string is cut so that the beam is no longer in equilibrium. Calculate the magnitude of the resultant moment about the pivot acting on the beam immediately after the string is cut.

$$\text{resultant moment} = \dots\dots\dots \text{ Nm [1]}$$

- (c) The beam in (b) rotates when the string is cut and the small block of weight 2.4 N is projected through the air. Fig. 2.2 shows the last part of the path of the block before it hits the ground at point Y.

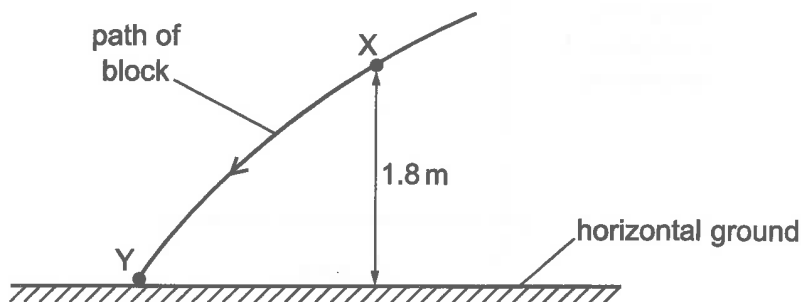


Fig. 2.2 (not to scale)

At point X on the path, the block has a speed of 3.4 ms^{-1} and is at a height of 1.8 m above the horizontal ground. Air resistance is negligible.

- (i) Calculate the decrease in the gravitational potential energy of the block for its movement from X to Y.

decrease in gravitational potential energy = J [2]

- (ii) Use your answer to (c)(i) and conservation of energy to determine the kinetic energy of the block at Y.

kinetic energy = J [3]

- (iii) State the variation, if any, in the direction of the acceleration of the block as it moves from X to Y.

..... [1]

- (iv) The block passes point X at time t_X and arrives at point Y at time t_Y .

On Fig. 2.3, sketch a graph to show the variation of the magnitude of the horizontal component of the velocity of the block with time from t_X to t_Y .

Numerical values are not required.

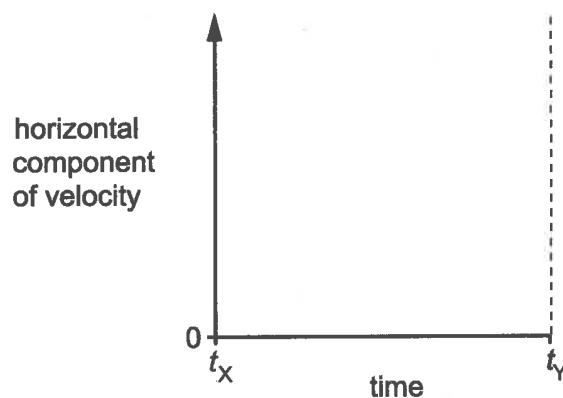


Fig. 2.3

[1]

28. M/J 23/P21/Q4

booklet (scan + OCR)

A beaker in air contains a liquid. The base of the beaker is in contact with the liquid and has area A , as shown in Fig. 4.1.

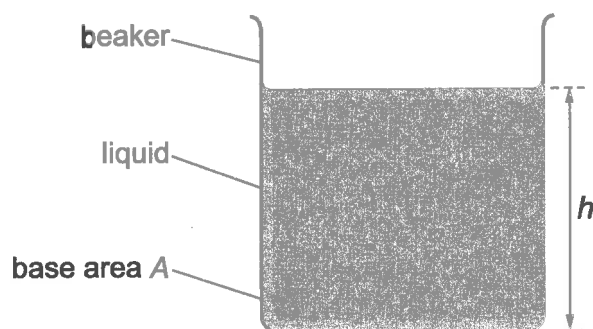


Fig. 4.1

The liquid has density ρ and fills the beaker to a depth h .

(a) By using the definitions of pressure and density, show that

$$p = \rho gh$$

where p is the pressure due to the liquid that is exerted on the base of the beaker and g is the acceleration of free fall.

[3]

(b) Suggest why the equation in (a) does not give the total pressure on the base of the beaker.

..... [1]

- (c) Fig. 4.2 shows the variation of the total pressure inside the liquid with depth x below the surface.

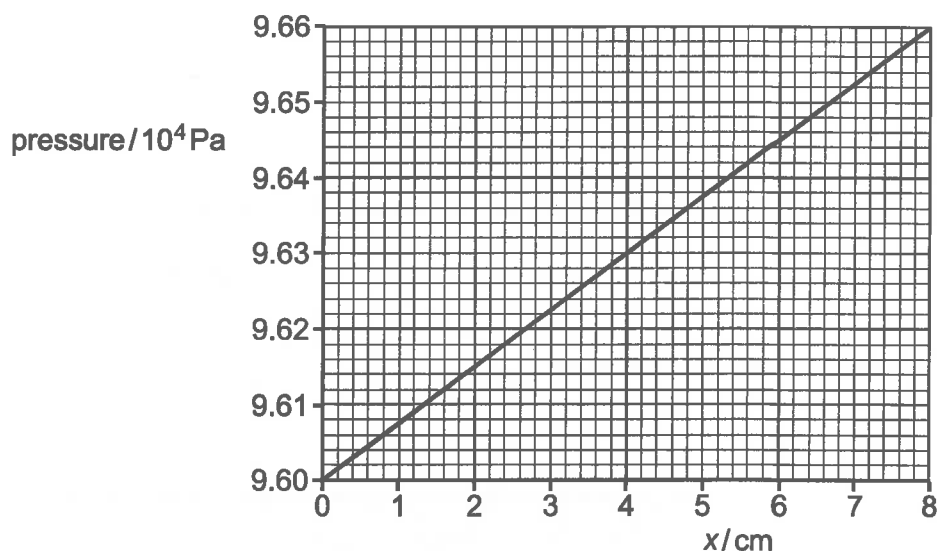


Fig. 4.2

Determine the density of the liquid.

density = kg m^{-3} [2]

- (d) A solid cylinder is held stationary by a wire so that the base of the cylinder is level with the surface of the liquid, as shown in Fig. 4.3.

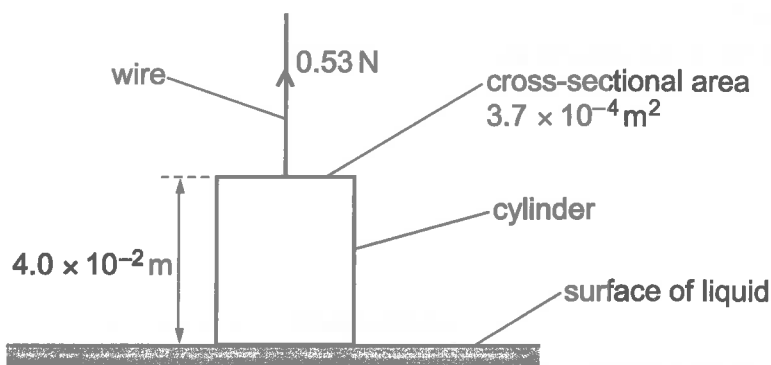


Fig. 4.3 (not to scale)

The cylinder has length $4.0 \times 10^{-2} \text{ m}$ and cross-sectional area $3.7 \times 10^{-4} \text{ m}^2$. The tension in the wire is 0.53 N .

The cylinder is now lowered and then held stationary by the wire so that the top of the cylinder is level with the surface of the liquid.

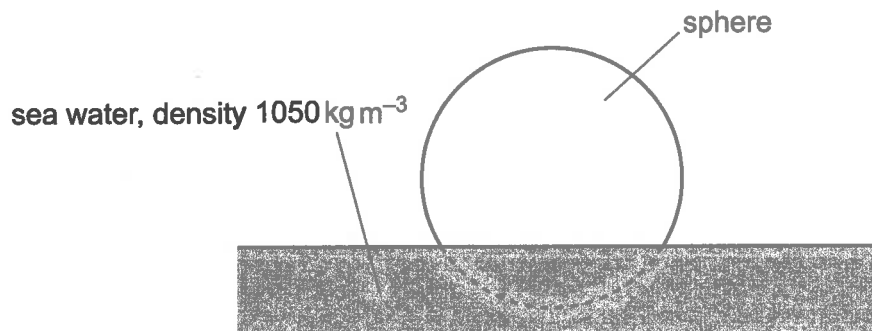
Calculate the new tension in the wire.

tension = N [2]

29. M/J 23/P23/Q2

booklet (scan + OCR)

A sphere floats in equilibrium on the surface of sea water of density 1050 kg m^{-3} , as shown in Fig. 2.1.

**Fig. 2.1**

- (a) 21% of the volume of the sphere is below the surface of the water.
Calculate the density of the sphere.

density = kg m^{-3} [2]

- (b) The sphere is now held so that its entire volume is below the surface of the water. The sphere is then released.
(i) Calculate the initial acceleration of the sphere.

acceleration = ms^{-2} [3]

- (ii) The sphere accelerates upwards but remains entirely below the surface of the water.
State and explain what happens to the acceleration of the sphere as its velocity begins to increase.

.....
.....
..... [3]

30. O/N 22/P22/Q2

booklet (scan + OCR)

A spherical balloon is filled with a fixed mass of gas. A small block is connected by a string to the balloon, as shown in Fig. 2.1.

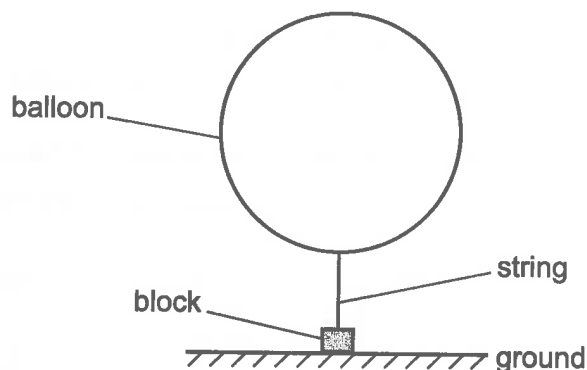


Fig. 2.1 (not to scale)

The block is held on the ground by an external force so that the string is vertical. The density of the air surrounding the balloon is 1.2 kg m^{-3} . The upthrust acting on the balloon is 0.071 N . The upthrust acting on the string and block is negligible.

(a) By using Archimedes' principle, calculate the radius r of the balloon.

$$r = \dots\dots\dots \text{ m [2]}$$

(b) The total weight of the balloon, string and block is 0.053 N .

The external force holding the block on the ground is removed so that the released block is lifted vertically upwards by the balloon.

Calculate the acceleration of the block immediately after it is released.

$$\text{acceleration} = \dots\dots\dots \text{ ms}^{-2} \text{ [3]}$$

(c) The balloon continues to lift the block. The string breaks as the block is moving vertically upwards with a speed of 1.4 m s^{-1} . After the string breaks, the detached block briefly continues moving upwards before falling vertically downwards to the ground. The block hits the ground with a speed of 3.6 m s^{-1} .

Assume that the air resistance on the block is negligible.

(i) By considering the motion of the block after the string breaks, calculate the height of the block above the ground when the string breaks.

height = m [2]

- (ii) The string breaks at time $t = 0$ and the block hits the ground at time $t = T$.

On Fig. 2.2, sketch a graph to show the variation of the velocity v of the block with time t from $t = 0$ to $t = T$.

Numerical values of t are not required. Assume that v is positive in the upward direction.

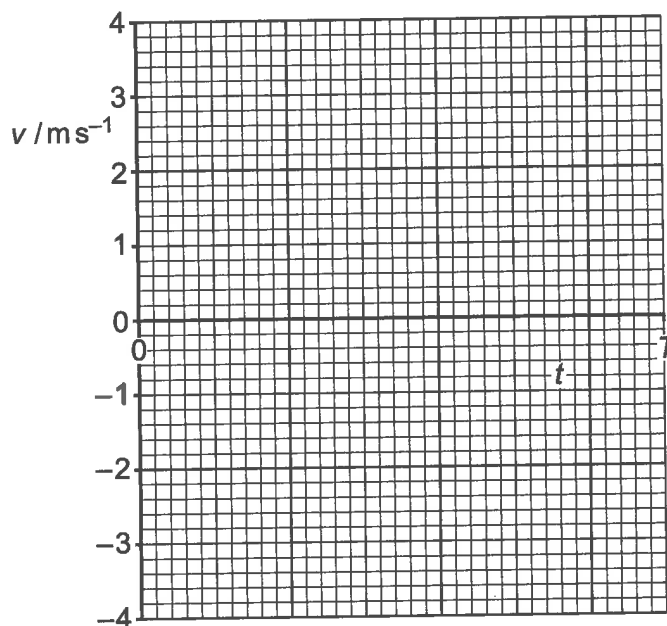


Fig. 2.2

[2]

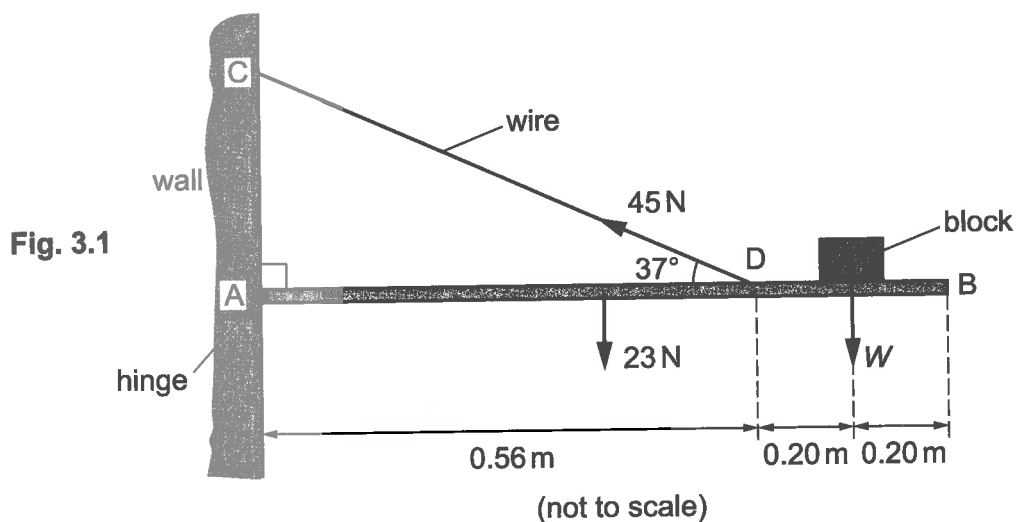
31. O/N 22/P22/Q3

booklet (scan + OCR)

- (a) State what is meant by the centre of gravity of an object.

..... [1]

- (b) A uniform beam AB is attached by a frictionless hinge to a vertical wall at end A. The beam is held so that it is horizontal by a metal wire CD, as shown in Fig. 3.1.



The beam is of length 0.96 m and weight 23 N. A block of weight W rests on the beam at a distance of 0.20 m from end B. The wire is attached to the beam at point D which is a distance of 0.40 m from end B. The wire exerts a force on the beam of 45 N at an angle of 37° to the horizontal. The beam is in equilibrium.

- (i) Calculate the vertical component of the force exerted by the wire on the beam.

vertical component of the force = N [1]

- (ii) By taking moments about A, calculate the weight W of the block.

$W =$ N [3]

- (iii) The hinge exerts a force on the beam at end A.
Calculate the horizontal component of this force.

horizontal component of force = N [1]

- (iv) The block is now placed closer to point D on the beam.
State whether this change will increase, decrease or have no effect on the tension in the wire.

..... [1]

- (v) The stress in the wire is 5.3×10^7 Pa. The wire is now replaced by a second wire that has a radius which is three times greater than that of the original wire. The tension in the wire is unchanged.
Calculate the stress in the second wire.

stress = Pa [2]

32. O/N 22/P23/Q3

booklet (scan + OCR)

- (a) State the principle of moments.

.....

..... [2]

- (b) A hollow plastic sphere is attached at one end of a bar. The sphere is partially submerged in water and the bar is attached to a fixed vertical support by a pivot P, as shown in Fig. 3.1.

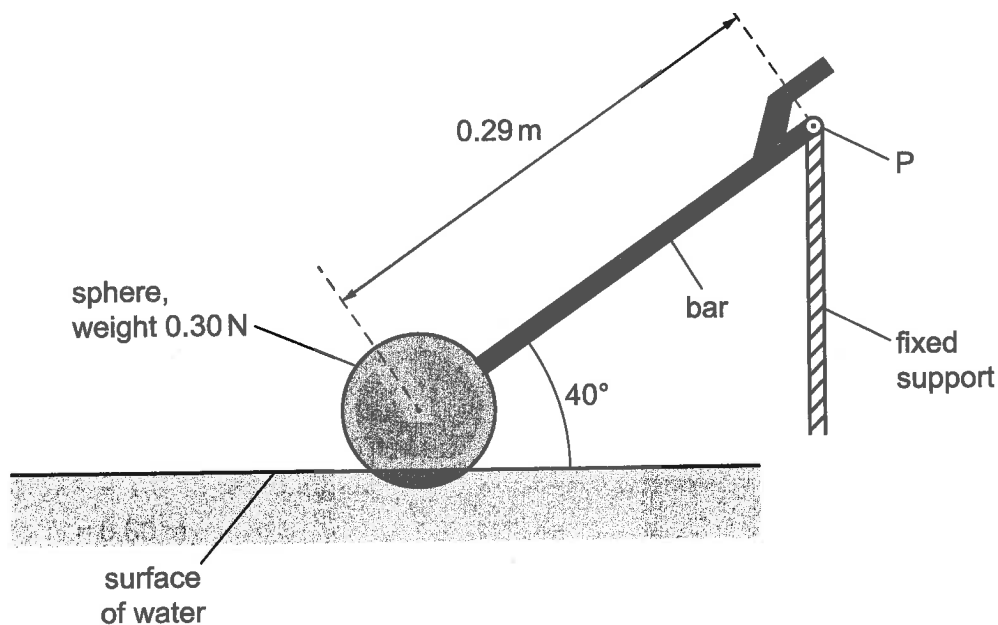


Fig. 3.1 (not to scale)

The sphere has weight 0.30 N. The distance from P to the centre of gravity of the sphere is 0.29 m. Assume that the weight of the bar is negligible. Calculate the moment of the weight of the sphere about P.

moment = N m [2]

- (c) The system shown in Fig. 3.1 is part of a mechanism that controls the amount of water in a tank.

Water enters the tank and causes the sphere to rise. This results in the bar becoming horizontal. Fig. 3.2 shows the system in its new position.

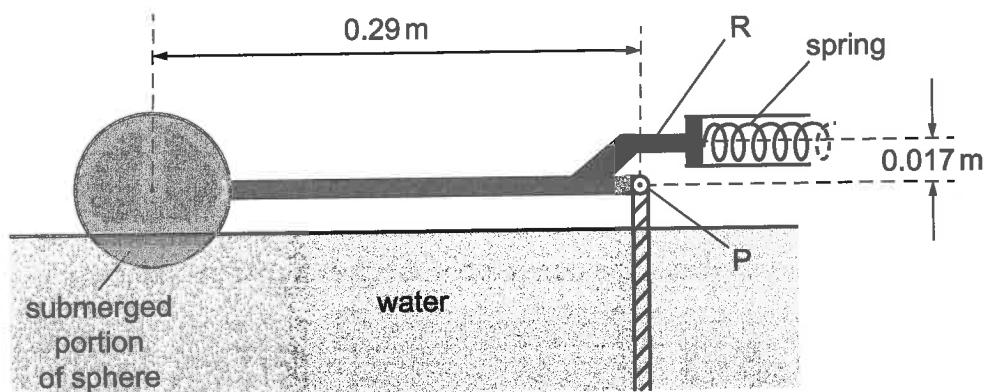


Fig. 3.2 (not to scale)

In this position the rod R exerts a force to compress a horizontal spring that controls the water supply to the tank. R is positioned at a perpendicular distance of 0.017 m above P. The variation of the force F applied to the spring with compression x of the spring is shown in Fig. 3.3.

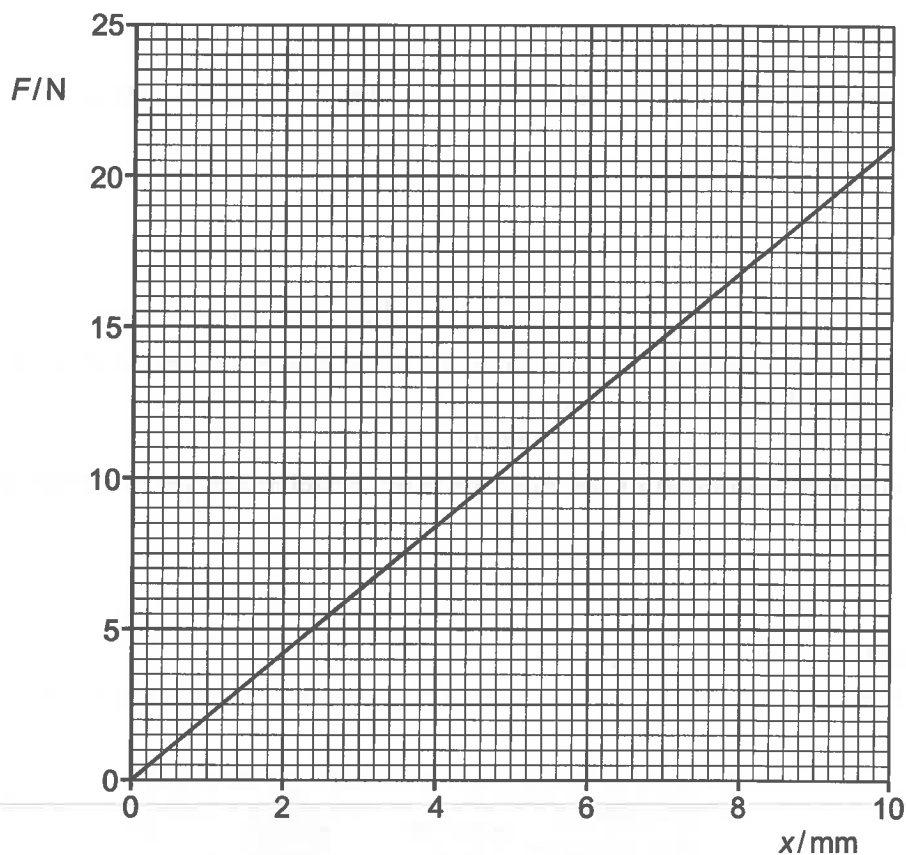


Fig. 3.3

- (i) Use Fig. 3.3 to calculate the spring constant k of the spring.

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

- (ii) At the position shown in Fig. 3.2, the system is stationary and in equilibrium. The radius of the sphere is 0.0480 m and 26.0% of the volume of the sphere is submerged. The density of water is $1.00 \times 10^3 \text{ kg m}^{-3}$. Show that the upthrust on the sphere is 1.18 N.

[2]

- (iii) By taking moments about P, determine the force exerted on the spring by the rod R.

force = N [2]

- (iv) Calculate the elastic potential energy E_p of the compressed spring.

E_p = J [2]

- (d) When the sphere moves from the position shown in Fig. 3.1 to the position shown in Fig. 3.2, the upthrust on the sphere does work.

Assume that resistive forces are negligible.

Explain why the work done by the upthrust is not equal to the gain in elastic potential energy of the spring.

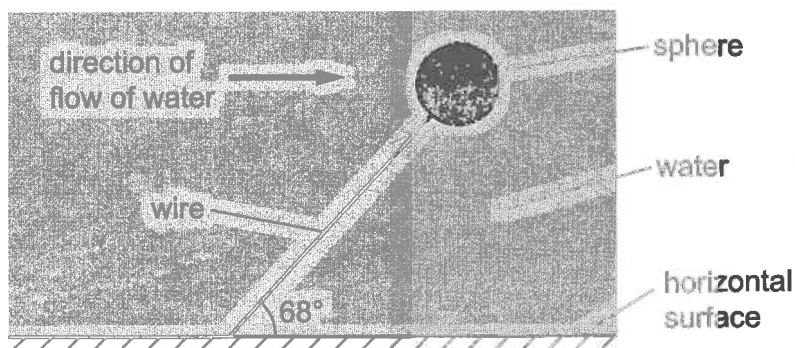
..... [1]

33. M/J 22/P22/Q2

booklet (scan + OCR)

A sphere is attached by a metal wire to the horizontal surface at the bottom of a river, as shown in Fig. 2.1.

Fig. 2.1 (not to scale)



The sphere is fully submerged and in equilibrium, with the wire at an angle of 68° to the horizontal surface. The weight of the sphere is 32 N. The upthrust acting on the sphere is 280 N. The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Assume that the force on the sphere due to the water flow is in a horizontal direction.

- (a) By considering the components of force in the vertical direction, determine the tension in the wire.

tension = N [2]

(b) For the sphere, calculate:

(i) the volume

volume = m^3 [1]

(ii) the density.

density = kg m^{-3} [2]

(c) The centre of the sphere is initially at a height of 6.2m above the horizontal surface. The speed of the water then increases, causing the sphere to move to a different position. This movement of the sphere causes its gravitational potential energy to decrease by 77 J.

Calculate the final height of the centre of the sphere above the horizontal surface.

height = m [3]

(d) The extension of the wire increases when the sphere changes position as described in (c). The wire obeys Hooke's law.

(i) State a symbol equation that gives the relationship between the tension T in the wire and its extension x . Identify any other symbol that you use.

..... [1]

(ii) Before the sphere changed position, the initial elastic potential energy of the wire was 0.65 J. The change in position of the sphere causes the extension of the wire to double. Calculate the final elastic potential energy of the wire after the sphere has changed position.

final elastic potential energy = J [2]

34. M/J 22/P21/Q2

booklet (scan + OCR)

(a) State what is meant by the centre of gravity of an object.

..... [1]

- (b) A non-uniform rod XY is pivoted at point P, as shown in Fig. 2.1.

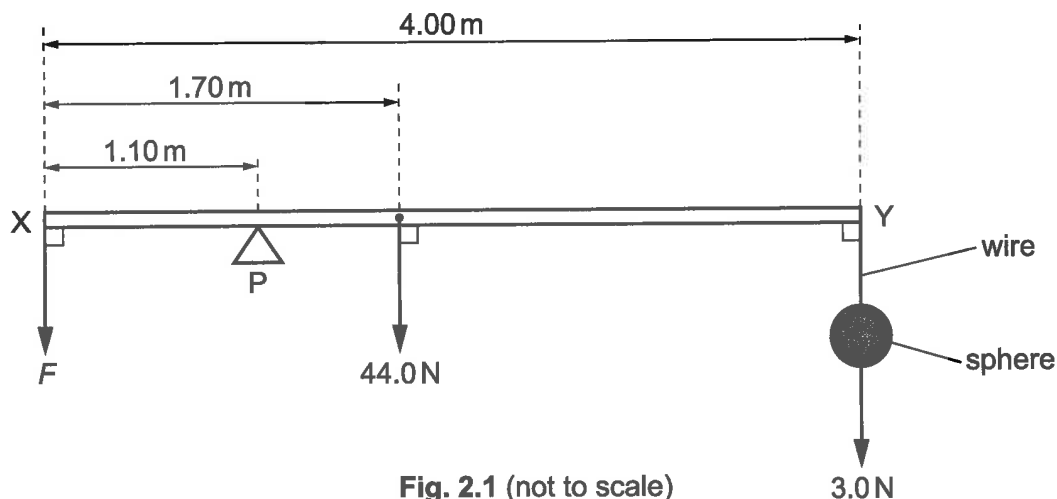


Fig. 2.1 (not to scale)

The rod has length 4.00 m and weight 44.0 N . The centre of gravity of the rod is 1.70 m from end X of the rod. Point P is 1.10 m from end X.

A sphere hangs by a wire from end Y of the rod. The weight of the sphere is 3.0 N . The weight of the wire is negligible.

A force F is applied vertically downwards at end X so that the horizontal rod is in equilibrium.

- (i) By taking moments about P, calculate F .

$$F = \dots\dots\dots \text{ N [3]}$$

- (ii) Calculate the force exerted on the rod by the pivot.

$$\text{force} = \dots\dots\dots \text{ N [1]}$$

- (c) The sphere in (b) is now immersed in a liquid in a container, as shown in Fig. 2.2.

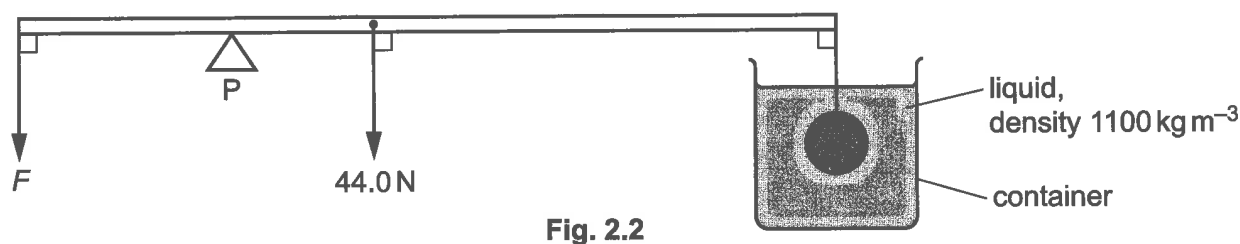


Fig. 2.2

The density of the liquid is 1100 kg m^{-3} . The upthrust acting on the sphere due to the liquid is 2.5 N . The magnitude of F is unchanged so that the horizontal rod is **not** in equilibrium.

- (i) Use Archimedes' principle to determine the radius r of the sphere.

$$r = \dots\dots\dots \text{ m [3]}$$

- (ii) Calculate the magnitude and direction of the resultant moment of the forces on the rod about P.

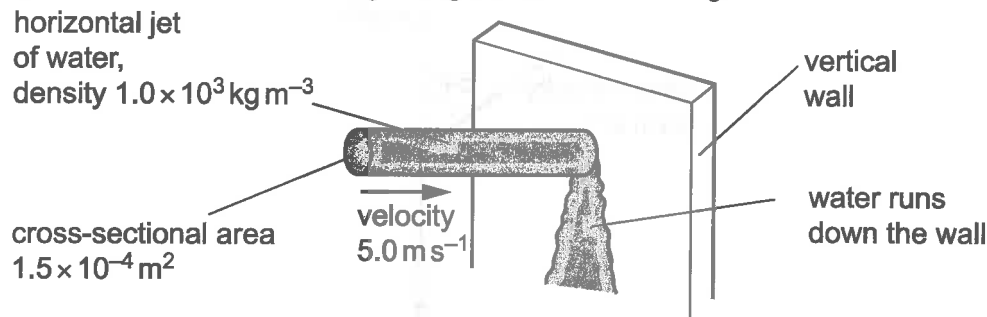
$$\text{magnitude of resultant moment} = \dots\dots\dots \text{ N m}$$

$$\text{direction of resultant moment} \dots\dots\dots [2]$$

35. MAR 22/P22/Q3

booklet (scan + OCR)

A jet of water hits a vertical wall at right angles, as shown in Fig. 3.1.

**Fig. 3.1** (not to scale)

The water hits the vertical wall with a velocity of 5.0 m s^{-1} in a horizontal direction. The cross-sectional area of the jet is $1.5 \times 10^{-4} \text{ m}^2$. The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$. The water runs down the wall after hitting it.

(a) Show that, over a time of 1.6 s, the mass of water hitting the wall is 1.2 kg.

[2]

(b) Calculate:

- (i) the decrease in the horizontal momentum of the mass of water in (a) due to hitting the wall

decrease in momentum = N s [1]

- (ii) the magnitude of the horizontal force exerted on the water by the wall.

force = N [1]

(c) State and explain the magnitude of the horizontal force exerted on the wall by the water.

..... [1]

(d) Calculate the pressure exerted on the wall by the water.

pressure = Pa [2]

36. O/N 21/P23/Q3

booklet (scan + OCR)

(a) Define *velocity*.

..... [1]

(b) A remote-controlled toy aircraft is flying horizontally in a wind. Fig. 3.1 shows the velocity vectors, to scale, of the wind and of the aircraft in still air.

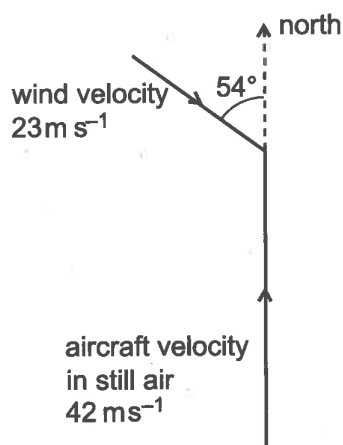


Fig. 3.1

The velocity of the aircraft in still air is 42 m s^{-1} to the north. The velocity of the wind is 23 m s^{-1} in a direction of 54° east of south.

Determine the magnitude of the resultant velocity of the aircraft.

magnitude of velocity = ms^{-1} [2]

(c) The engine of the aircraft in (b) stops. The aircraft then glides towards the ground with a constant velocity at an angle θ to the horizontal, as illustrated in Fig. 3.2.

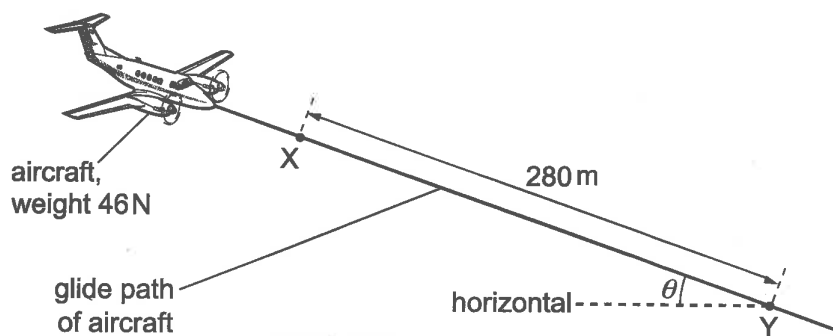


Fig. 3.2 (not to scale)

The aircraft has a weight of 46 N and travels a distance of 280 m from point X to point Y. The change in gravitational potential energy of the aircraft for its movement from X to Y is 6100 J. Assume that there is now no wind.

- (i) Calculate angle θ .

$$\theta = \dots\dots\dots^\circ \quad [3]$$

- (ii) Calculate the magnitude of the force acting on the aircraft due to air resistance.

$$\text{force} = \dots\dots\dots \text{ N} \quad [2]$$

- (d) The aircraft in (c) travels from X to Y in a time of 14 s. Fig. 3.3 shows that, as the aircraft travels from X to Y, it moves directly towards an observer who is standing on the ground.

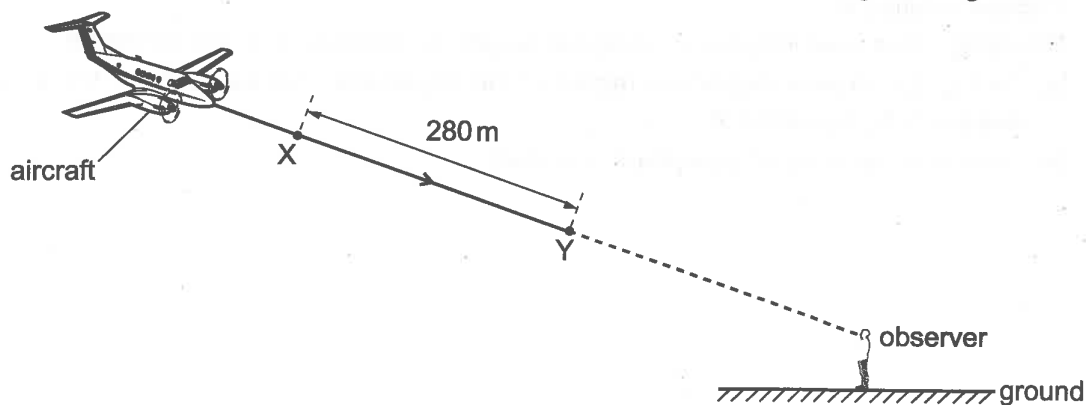


Fig. 3.3 (not to scale)

The aircraft emits sound as it travels from X to Y. The observer hears sound of frequency 450 Hz. The speed of the sound in the air is 340 ms^{-1} .

Calculate the frequency of the sound that is emitted by the aircraft.

frequency = Hz [3]

37. M/J 21/P21/Q3

booklet (scan + OCR)

A pendulum consists of a solid sphere suspended by a string from a fixed point P, as shown in Fig. 3.1.

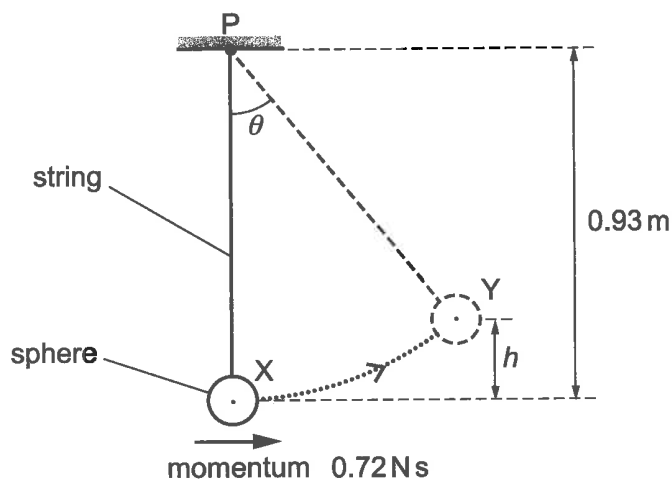


Fig. 3.1 (not to scale)

The sphere swings from side to side. At one instant the sphere is at its lowest position X, where it has kinetic energy 0.86 J and momentum 0.72 N s in a horizontal direction. A short time later the sphere is at position Y, where it is momentarily stationary at a maximum vertical height h above position X.

The string has a fixed length and negligible weight. Air resistance is also negligible.

- (a) On Fig. 3.1, draw a solid line to represent the displacement of the centre of the sphere at position Y from position X. [1]
- (b) Show that the mass of the sphere is 0.30 kg. [3]

(c) Calculate height h .

$$h = \dots\dots\dots\text{m} \quad [2]$$

(d) The distance between point P and the centre of the sphere is 0.93m. When the sphere is at position Y, the string is at an angle θ to the vertical.
Show that θ is 47° .

[1]

(e) For the sphere at position Y, calculate the moment of its weight about point P.

$$\text{moment} = \dots\dots\dots\text{Nm} \quad [2]$$

(f) State and explain whether the sphere is in equilibrium when it is stationary at position Y.

..... [1]

38. M/J 21/P23/Q1

booklet (scan + OCR)

(a) A property of a vector quantity, that is not a property of a scalar quantity, is direction.
For example, velocity has direction but speed does not.

(i) State **two** other scalar quantities and **two** other vector quantities.

scalar quantities: and

vector quantities: and [2]

(ii) State **two** properties that are possessed by both scalar and vector physical quantities.

1.

2. [2]

(b) A ship at sea is travelling with a velocity of 13 m s^{-1} in a direction 35° east of north in still water, as shown in Fig. 1.1.

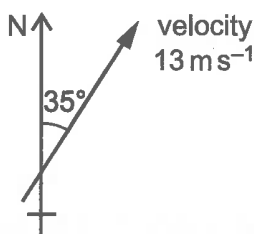
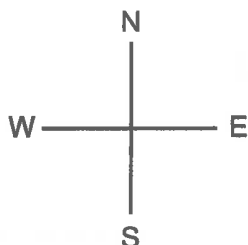


Fig. 1.1

- (i) Determine the magnitudes of the components of the velocity of the ship in the north and the east directions.

north component of velocity = ms^{-1}

east component of velocity = ms^{-1} [2]

- (ii) The ship now experiences a tidal current. The water in the sea moves with a velocity of 2.7 ms^{-1} to the west.

Calculate the resultant velocity component of the ship in the east direction.

resultant east component of velocity = ms^{-1} [1]

- (iii) Use your answers in (b)(i) and (b)(ii) to determine the magnitude of the resultant velocity of the ship.

magnitude of resultant velocity = ms^{-1} [2]

- (iv) Use your answers in (b)(i) and (b)(ii) to determine the angle between north and the resultant velocity of the ship.

angle =° [2]

39. O/N 20/P22/Q2

booklet (scan + OCR)

- (a) A cylinder is suspended from the end of a string. The cylinder is stationary in water with the axis of the cylinder vertical, as shown in Fig. 2.1.

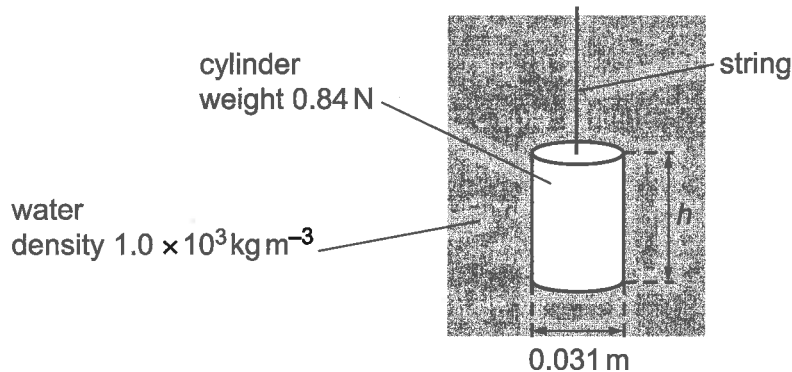


Fig. 2.1 (not to scale)

The cylinder has weight 0.84 N , height h and a circular cross-section of diameter 0.031 m . The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$. The difference between the pressures on the top and bottom faces of the cylinder is 520 Pa .

(i) Calculate the height h of the cylinder.

$$h = \dots\dots\dots \text{ m } [2]$$

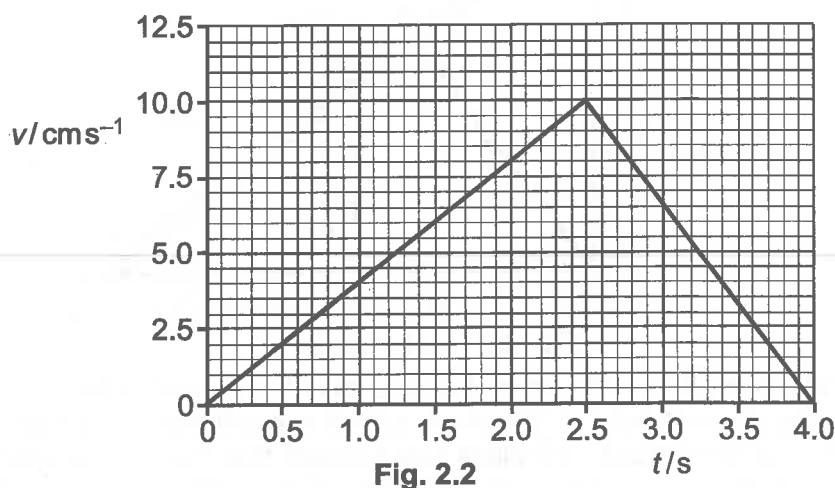
(ii) Show that the upthrust acting on the cylinder is 0.39N.

[2]

(iii) Calculate the tension T in the string.

$$T = \dots\dots\dots \text{ N } [1]$$

(b) The string is now used to move the cylinder in (a) vertically upwards through the water. The variation with time t of the velocity v of the cylinder is shown in Fig. 2.2.



(i) Use Fig. 2.2 to determine the acceleration of the cylinder at time $t = 2.0$ s.

$$\text{acceleration} = \dots\dots\dots \text{ ms}^{-2} [2]$$

(ii) The top face of the cylinder is at a depth of 0.32m below the surface of the water at time $t = 0$.

Use Fig. 2.2 to determine the depth of the top face below the surface of the water at time $t = 4.0$ s.

$$\text{depth} = \dots\dots\dots \text{ m } [2]$$

(c) The cylinder in (b) is released from the string at time $t = 4.0$ s. The cylinder falls, from rest, vertically downwards through the water. Assume that the upthrust acting on the cylinder remains constant as it falls.

(i) State the name of the force that acts on the cylinder when it is moving and does not act on the cylinder when it is stationary.

..... [1]

(ii) State and explain the variation, if any, of the acceleration of the cylinder as it falls downwards through the water.

.....

..... [2]

40. O/N 20/P22/Q4

booklet (scan + OCR)

A rigid plank is used to make a ramp between two different horizontal levels of ground, as shown in Fig. 4.1.

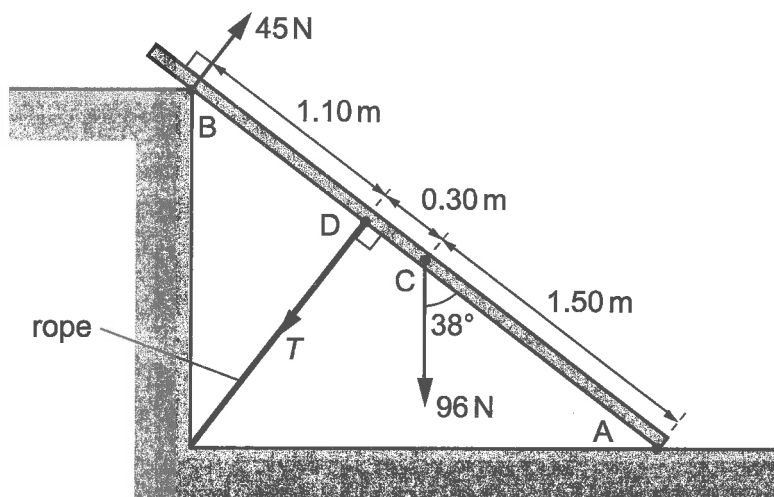


Fig. 4.1 (not to scale)

Point A at one end of the plank rests on the lower level of the ground. A force acts on, and is perpendicular to, the plank at point B. The plank is held in equilibrium by a rope that connects point D on the plank to the ground. The plank has a weight that may be considered to act from its centre of gravity C. The rope is perpendicular to the plank and has tension T . The plank is at an angle of 38° to the vertical.

The forces and the distances along the plank of points A, B, C and D are shown in Fig. 4.1.

(a) Show that the component of the weight that is perpendicular to the plank is 59 N.

[1]

(b) By taking moments about end A of the plank, calculate the tension T .

$T = \dots\dots\dots$ N [3]

41. O/N 20/P21/Q1

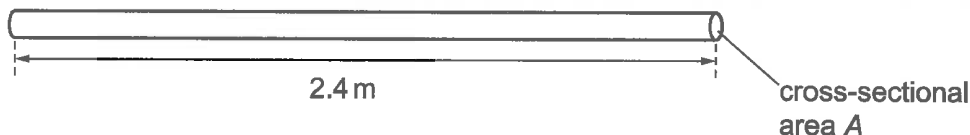
booklet (scan + OCR)

(a) (i) Define the *moment* of a force about a point.

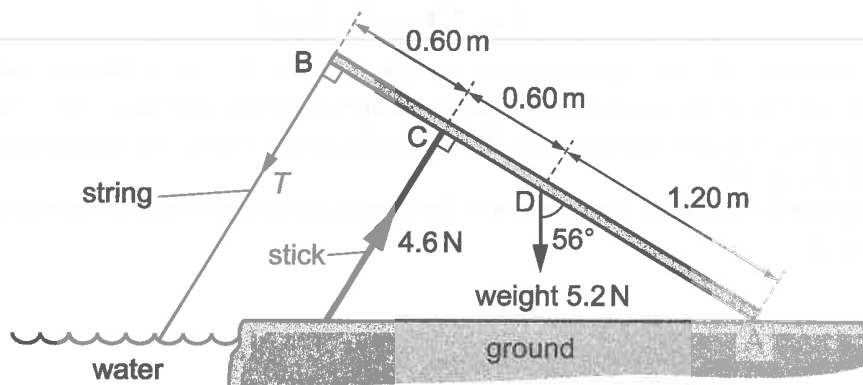
..... [1]

(ii) Determine the SI base units of the moment of a force.

base units [1]

(b) A uniform rigid rod of length 2.4 m is shown in Fig. 1.1.**Fig. 1.1**

The rod has a weight of 5.2 N and is made of wood of density 790 kg m^{-3} . Calculate the cross-sectional area A , in mm^2 , of the rod.

 $A = \dots\dots\dots \text{mm}^2$ [3]**(c)** A fishing rod AB, made from the rod in (b), is shown in Fig. 1.2.**Fig. 1.2** (not to scale)

End A of the rod rests on the ground and a string is attached to the other end B. A support stick exerts a force perpendicular to the rod at point C. The weight of the rod acts at point D. The tension T in the string is in a direction perpendicular to the rod. The rod is in equilibrium and inclined at an angle of 56° to the vertical.

The forces and the distances along the rod of points A, B, C and D are shown in Fig. 1.2.

(i) Show that the component of the weight that is perpendicular to the rod is 4.3 N.

[1]

(ii) By taking moments about end A of the rod, calculate the tension T .

$$T = \dots\dots\dots \text{ N [3]}$$

42. O/N 20/P23/Q2

booklet (scan + OCR)

(a) State what is meant by the *centre of gravity* of a body.

.....
 [2]

(b) A uniform wooden post AB of weight 45 N stands in equilibrium on hard ground, as shown in Fig. 2.1.

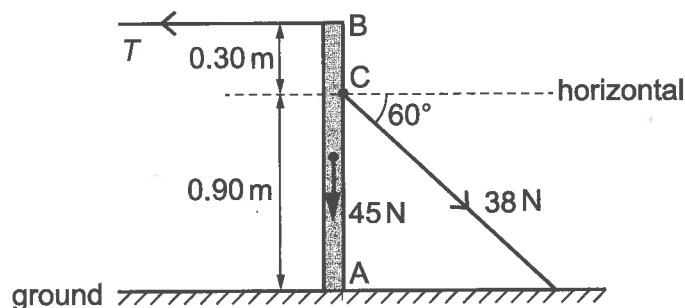


Fig. 2.1 (not to scale)

End A of the vertical post is supported by the ground. A horizontal wire with tension T is attached to end B of the post. Another wire, attached to the post at point C, is at an angle of 60° to the horizontal and has tension 38 N. The distances along the post of points A, B and C are shown in Fig. 2.1.

(i) Calculate the horizontal component of the force exerted on the post by the wire connected to point C.

$$\text{horizontal component of force} = \dots\dots\dots \text{ N [1]}$$

(ii) By considering moments about end A, determine the tension T .

$$T = \dots\dots\dots \text{ N [2]}$$

(iii) Calculate the vertical component of the force exerted on the post at end A.

$$\text{force} = \dots\dots\dots \text{ N [1]}$$

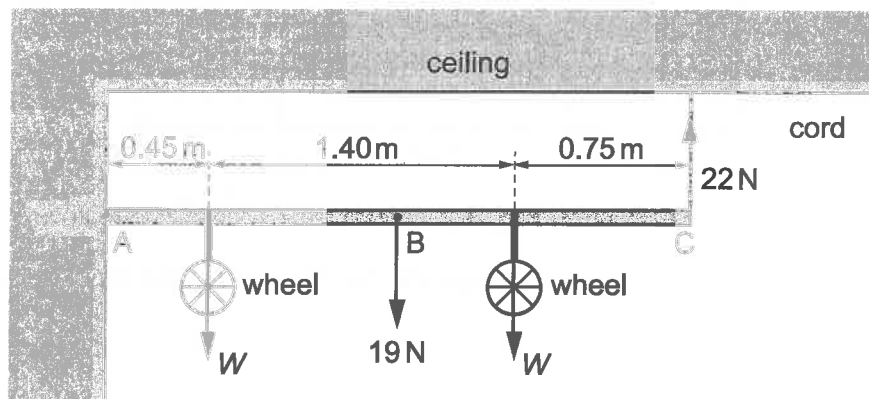
43. M/J 20/P23/Q3

booklet (scan + OCR)

(a) State the principle of moments.

.....

..... [2]

(b) In a bicycle shop, two wheels hang from a horizontal uniform rod AC, as shown in Fig. 3.1.**Fig. 3.1** (not to scale)

The rod has weight 19 N and is freely hinged to a wall at end A. The other end C of the rod is attached by a vertical elastic cord to the ceiling. The centre of gravity of the rod is at point B. The weight of each wheel is W and the tension in the cord is 22 N.

(i) By taking moments about end A, show that the weight W of each wheel is 14 N.

[2]

(ii) Determine the magnitude and the direction of the force acting on the rod at end A.

magnitude = N

direction [2]

- (c) The unstretched length of the cord in (b) is 0.25 m. The variation with length L of the tension F in the cord is shown in Fig. 3.2.

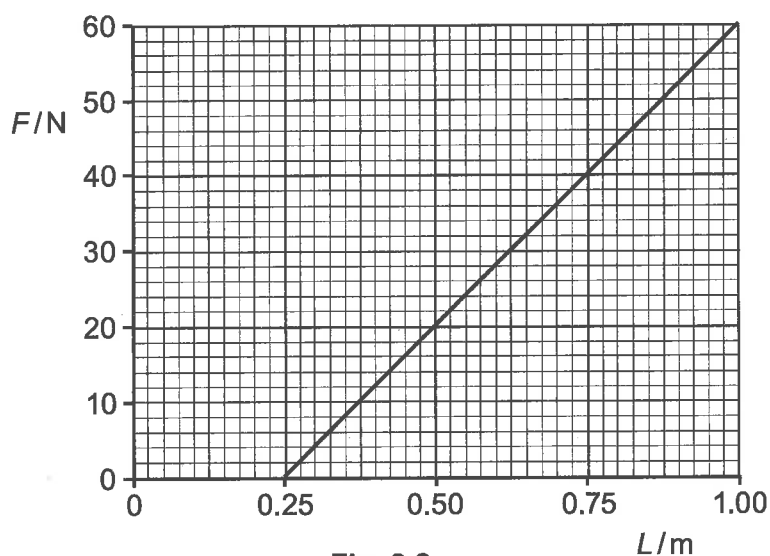


Fig. 3.2

- (i) State and explain whether Fig. 3.2 suggests that the cord obeys Hooke's law.

.....
 [2]

- (ii) Calculate the spring constant k of the cord.

$$k = \dots\dots\dots \text{N m}^{-1} \quad [2]$$

- (iii) On Fig. 3.2, shade the area that represents the work done to extend the cord when the tension is increased from $F = 0$ to $F = 40$ N. [1]

44. O/N 19/P22/Q4

booklet (scan + OCR)

- (a) A sphere in a liquid accelerates vertically downwards from rest. For the viscous force acting on the moving sphere, state:

(i) the direction

..... [1]

(ii) the variation, if any, in the magnitude.

..... [1]

- (b) A man of weight 750 N stands a distance of 3.6 m from end D of a horizontal uniform beam AD, as shown in Fig. 4.1.

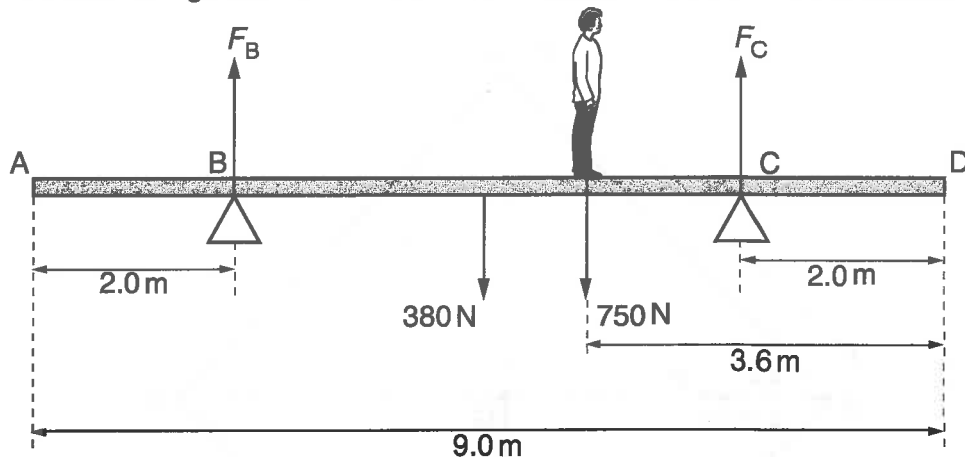


Fig. 4.1 (not to scale)

The beam has a weight of 380 N and a length of 9.0 m. The beam is supported by a vertical force F_B at pivot B and a vertical force F_C at pivot C. Pivot B is a distance of 2.0 m from end A and pivot C is a distance of 2.0 m from end D. The beam is in equilibrium.

- (i) State the principle of moments.

.....

..... [2]

- (ii) By using moments about pivot C, calculate F_B .

$F_B = \dots\dots\dots$ N [2]

- (iii) The man walks towards end D. The beam is about to tip when F_B becomes zero.

Determine the minimum distance x from end D that the man can stand without tipping the beam.

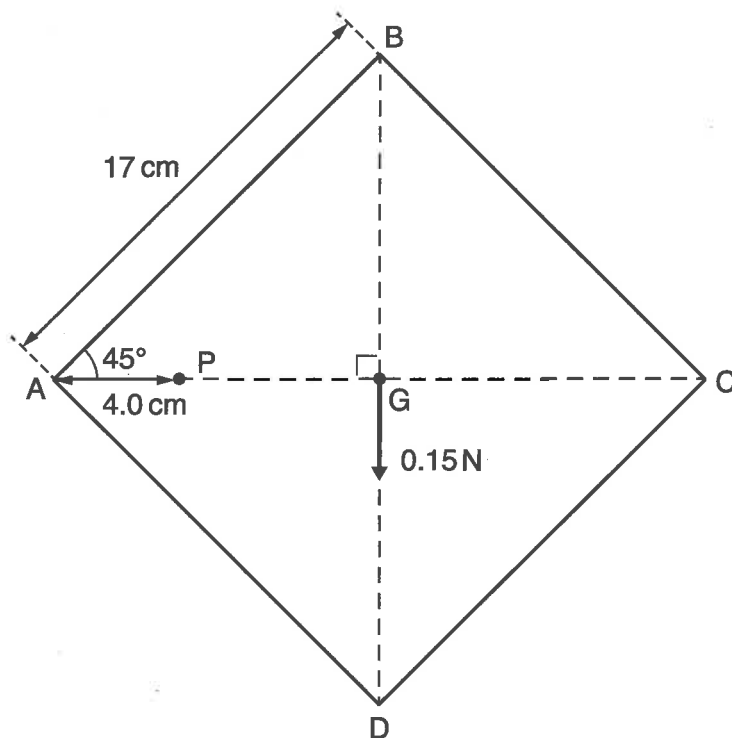
$x = \dots\dots\dots$ m [2]

45. O/N 19/P23/Q1

booklet (scan + OCR)

(a) Determine the SI base units of the moment of a force.

SI base units [1]

(b) A uniform square sheet of card ABCD is freely pivoted by a pin at a point P. The card is held in a vertical plane by an external force in the position shown in Fig. 1.1.**Fig. 1.1** (not to scale)

The card has weight 0.15 N which may be considered to act at the centre of gravity G. Each side of the card has length 17 cm. Point P lies on the horizontal line AC and is 4.0 cm from corner A. Line BD is vertical.

The card is released by removing the external force. The card then swings in a vertical plane until it comes to rest.

(i) Calculate the magnitude of the resultant moment about point P acting on the card immediately after it is released.

moment = Nm [2]

(ii) Explain why, when the card has come to rest, its centre of gravity is vertically below point P.

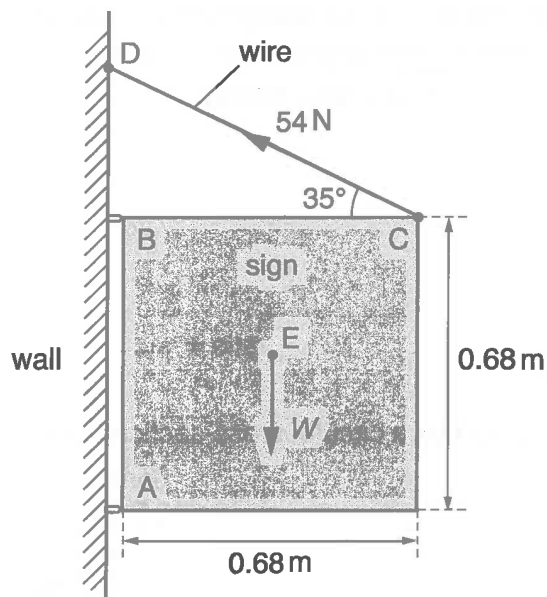
.....
 [2]

46. M/J 19/P22/Q3

booklet (scan + OCR)

(a) State what is meant by the *centre of gravity* of a body.

..... [1]

(b) A uniform square sign with sides of length 0.68 m is fixed at its corner points A and B to a wall. The sign is also supported by a wire CD, as shown in Fig. 3.1.**Fig. 3.1** (not to scale)

The sign has weight W and centre of gravity at point E. The sign is held in a vertical plane with side BC horizontal. The wire is at an angle of 35° to side BC. The tension in the wire is 54 N. The force exerted on the sign at B is only in the vertical direction.

(i) Calculate the vertical component of the tension in the wire.

vertical component of tension = N [1]

(ii) Explain why the force on the sign at B does not have a moment about point A.

..... [1]

(iii) By taking moments about point A, show that the weight W of the sign is 150 N.

[2]

(iv) Calculate the total vertical force exerted by the wall on the sign at points A and B.

total vertical force =N [1]

- (c) The sign in (b) is held together by nuts and bolts. One of the nuts falls vertically from rest through a distance of 4.8 m to the pavement below. The nut lands on the pavement with a speed of 9.2 ms^{-1} . Determine, for the nut falling from the sign to the pavement, the ratio
- $$\frac{\text{change in gravitational potential energy}}{\text{final kinetic energy}}.$$

ratio = [4]

47. M/J 19/P23/Q3

booklet (scan + OCR)

A cylindrical disc of mass 0.24 kg has a circular cross-sectional area A , as shown in Fig. 3.1.

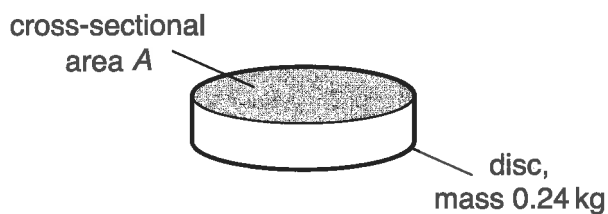


Fig. 3.1

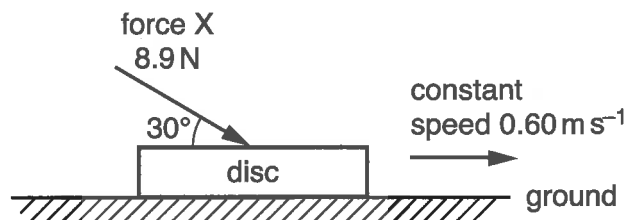


Fig. 3.2

The disc is on horizontal ground, as shown in Fig. 3.2. A force X of magnitude 8.9 N acts on the disc in a direction of 30° to the horizontal. The disc moves at a constant speed of 0.60 ms^{-1} along the ground.

- (a) Determine the rate of doing work on the disc by the force X .

rate of doing work = W [2]

- (b) The force X and the weight of the disc exert a combined pressure on the ground of 3500 Pa. Calculate the cross-sectional area A of the disc.

$A = \dots\dots\dots \text{m}^2$ [3]

- (c) Newton's third law describes how forces exist in pairs. One such pair of forces is the weight of the disc and another force Y . State:

- (i) the direction of force Y

..... [1]

- (ii) the name of the body on which force Y acts.

..... [1]

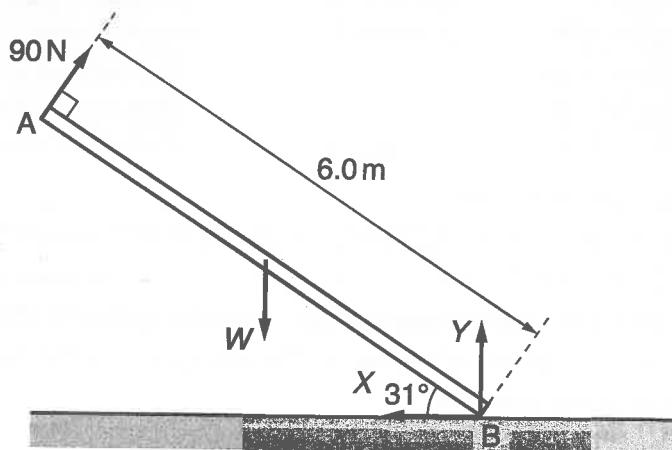
48. O/N 18/P22/Q2

booklet (scan + OCR)

- (a)** The kilogram, metre and second are all SI base units.
State two other SI base units.

1.
2. [2]

- (b)** A uniform beam AB of length 6.0 m is placed on a horizontal surface and then tilted at an angle of 31° to the horizontal, as shown in Fig. 2.1.

**Fig. 2.1** (not to scale)

The beam is held in equilibrium by four forces that all act in the same plane. A force of 90 N acts perpendicular to the beam at end A. The weight W of the beam acts at its centre of gravity. A vertical force Y and a horizontal force X both act at end B of the beam.

- (i)** State the name of force X .

..... [1]

- (ii)** By taking moments about end B, calculate the weight W of the beam.

$W =$ N [2]

- (iii)** Determine the magnitude of force X .

magnitude of force $X =$ N [1]

49. M/J 18/P22/Q5

booklet (scan + OCR)

A solid cylinder is lifted out of oil by a wire attached to a motor. Fig. 5.1 shows two different positions X and Y of the cylinder during the lifting process.

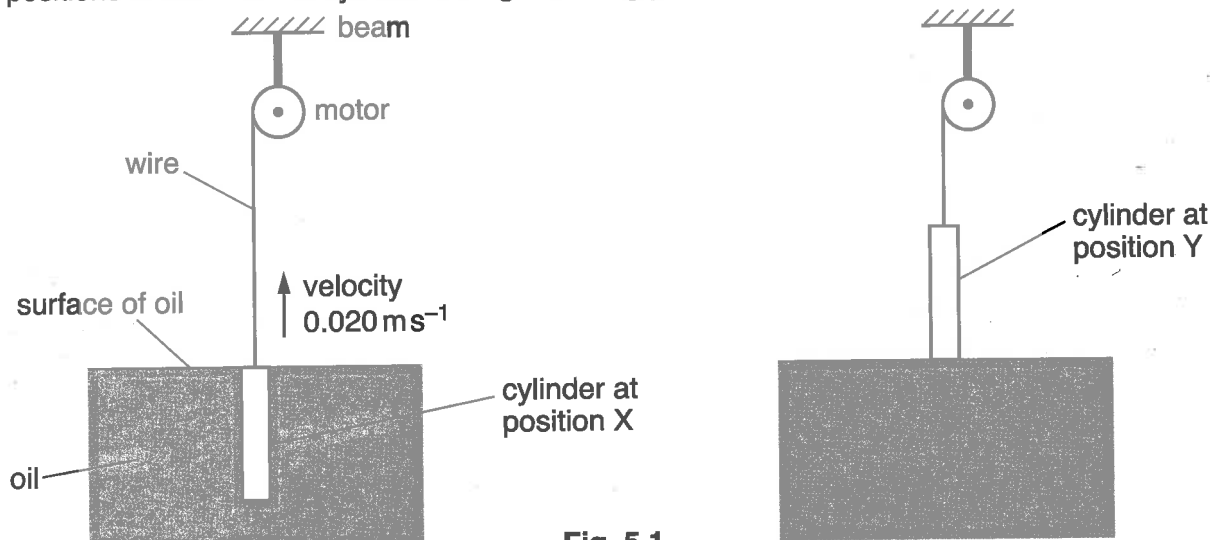


Fig. 5.1

The motor is fixed to an overhead beam.

The cylinder has cross-sectional area 0.018 m^2 , length 1.2 m and weight 560 N .

The density of the oil is 940 kg m^{-3} .

Throughout the lifting process, the cylinder moves vertically upwards with a constant velocity of 0.020 ms^{-1} . The viscous force of the oil acting on the cylinder is negligible.

(a) Calculate the density of the cylinder.

density = kg m^{-3} [2]

(b) For the cylinder at position X, show that the upthrust due to the oil is 200 N .

[2]

(c) Calculate, for the moving cylinder at position X,

(i) the tension in the wire,

tension = N [1]

(ii) the power output of the motor.

power = W [2]

(d) The cylinder is raised with constant velocity from position X to position Y.

- (i)** State and explain the variation, if any, of the power output of the motor as the cylinder is raised. Numerical values are not required.

.....

.....

..... [3]

- (ii)** The rate of energy output of the motor is less than the rate of increase of gravitational potential energy of the cylinder. Without calculation, explain this difference.

..... [1]

50. M/J 18/P23/Q2

booklet (scan + OCR)

(a) State what is meant by *work done*.

..... [1]

(b) A diver releases a solid sphere of radius 16 cm from the sea bed. The sphere moves vertically upwards towards the surface of the sea.

The weight of the sphere is 20 N. The upthrust acting on the sphere is 170 N. The upthrust remains constant as the sphere moves upwards.

(i) Calculate the density of the material of the sphere.density = kg m^{-3} [2]**(ii)** Briefly explain the origin of the upthrust acting on the sphere.

..... [1]

(iii) Calculate the acceleration of the sphere as it is released from rest.acceleration = ms^{-2} [2]**(iv)** The viscous (drag) force D acting on the sphere is given by

$$D = kr^2v^2$$

where r is the radius of the sphere and v is its speed.The constant k is equal to 810 kg m^{-3} .

Determine the constant (terminal) speed reached by the sphere.

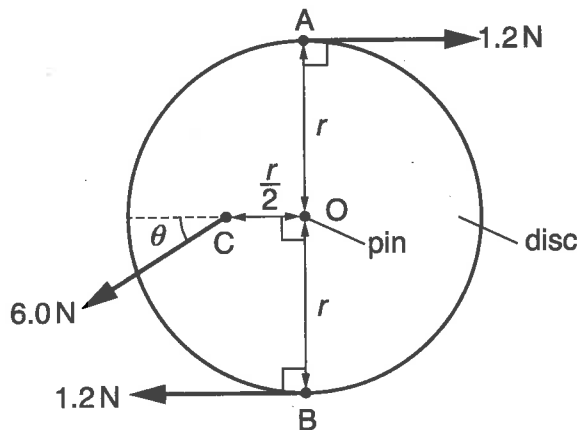
speed = ms^{-1} [3]**(v)** The diver releases a different sphere that moves with a constant speed of 6.30 ms^{-1} directly towards a stationary ship. The sphere emits sound of frequency 4850 Hz. The ship detects sound of frequency 4870 Hz as the sphere moves towards it. Determine, to three significant figures, the speed of the sound in the water.speed = ms^{-1} [2]

51. O/N 17/P22/Q2

booklet (scan + OCR)

(a) Define the *moment* of a force.

..... [1]

(b) A thin disc of radius r is supported at its centre O by a pin. The disc is supported so that it is vertical. Three forces act in the plane of the disc, as shown in Fig. 2.1.**Fig. 2.1**

Two horizontal and opposite forces, each of magnitude 1.2 N, act at points A and B on the edge of the disc. A force of 6.0 N, at an angle θ below the horizontal, acts on the midpoint C of a radial line of the disc, as shown in Fig. 2.1. The disc has negligible weight and is in equilibrium.

(i) State an expression, in terms of r , for the torque of the couple due to the forces at A and B acting on the disc.

..... [1]

(ii) Friction between the disc and the pin is negligible.
Determine the angle θ .

 $\theta =$ ° [2]

(iii) State the magnitude of the force of the pin on the disc.

force = N [1]

52. O/N 17/P23/Q2

booklet (scan + OCR)

A liquid of density ρ fills a container to a depth h , as shown in Fig. 2.1.

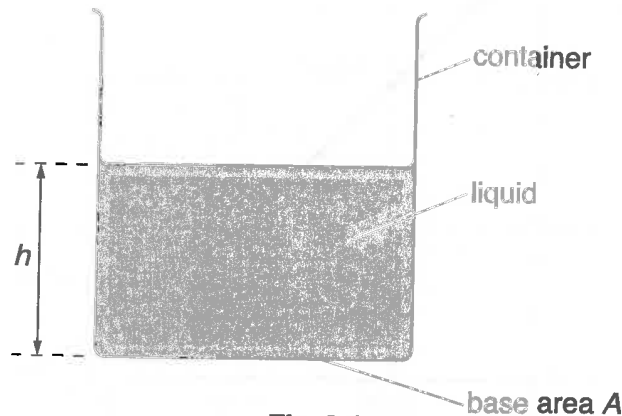


Fig. 2.1

The base of the container has area A .

(a) Derive, from the definitions of pressure and density, the equation

$$p = \rho gh$$

where p is the pressure exerted by the liquid on the base of the container and g is the acceleration of free fall.

[3]

(b) A small solid sphere falls with constant velocity through the liquid.

(i) State

1. the names of the three forces acting on the sphere,

.....

2. a word equation that relates the magnitudes of these forces.

..... [2]

(ii) State and explain the changes in energy that occur as the sphere falls.

.....

..... [2]

(c) The liquid in the container is liquid L. Liquid M is now added to the container. The two liquids do not mix. The total depth of the liquids is 0.17 m.

Fig. 2.2 shows how the pressure p inside the liquids varies with height x above the base of the container.

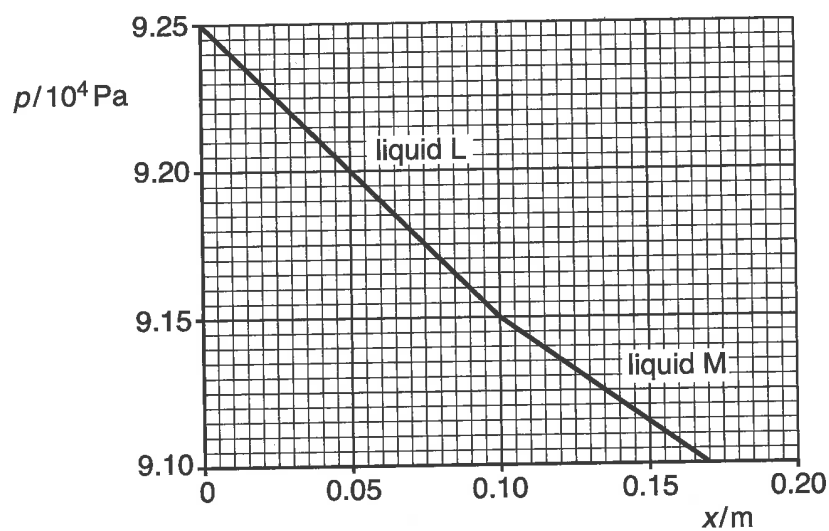


Fig. 2.2

Use Fig. 2.2 to

- (i) state the value of atmospheric pressure,

atmospheric pressure = Pa [1]

- (ii) determine the density of liquid M.

density = kg m^{-3} [2]

53. M/J 17/P21/Q3

booklet (scan + OCR)

- (a) A cylinder is made from a material of density 2.7 g cm^{-3} . The cylinder has diameter 2.4 cm and length 5.0 cm.

Show that the cylinder has weight 0.60 N.

[3]

(b) The cylinder in (a) is hung from the end A of a non-uniform bar AB, as shown in Fig. 3.1.

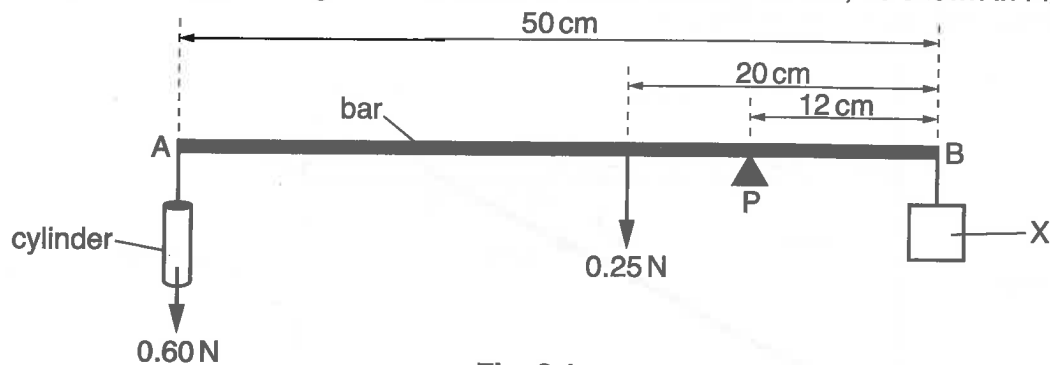


Fig. 3.1

The bar has length 50 cm and has weight 0.25 N. The centre of gravity of the bar is 20 cm from B. The bar is pivoted at P. The pivot is 12 cm from B.

An object X is hung from end B. The weight of X is adjusted until the bar is horizontal and in equilibrium.

(i) Explain what is meant by *centre of gravity*.

..... [1]

(ii) Calculate the weight of X.

weight of X = N [3]

(c) The cylinder is now immersed in water, as illustrated in Fig. 3.2.

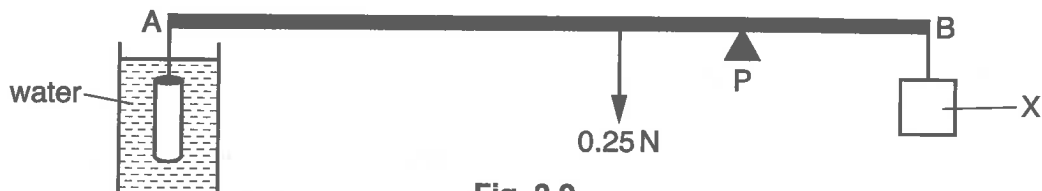


Fig. 3.2

An upthrust acts on the cylinder and the bar is not in equilibrium.

(i) Explain the origin of the upthrust.

..... [2]

(ii) Explain why the weight of X must be reduced in order to obtain equilibrium for AB.

..... [1]

54. O/N 16/P22/Q3

booklet (scan + OCR)

(a) State the two conditions for an object to be in equilibrium.

1.

2. [2]

- (b) A uniform beam AC is attached to a vertical wall at end A. The beam is held horizontal by a rigid bar BD, as shown in Fig. 3.1.

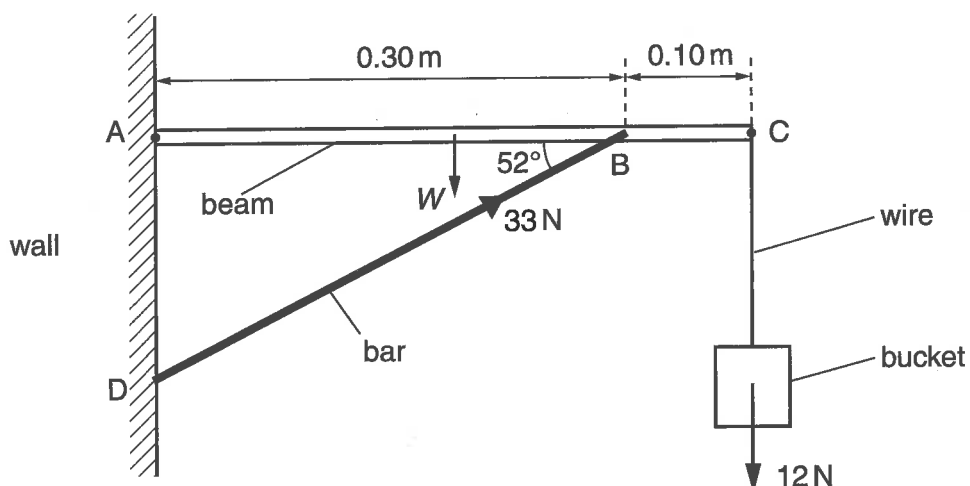


Fig. 3.1 (not to scale)

The beam is of length 0.40 m and weight W . An empty bucket of weight 12 N is suspended by a light metal wire from end C. The bar exerts a force on the beam of 33 N at 52° to the horizontal. The beam is in equilibrium.

- (i) Calculate the vertical component of the force exerted by the bar on the beam.

component of the force = N [1]

- (ii) By taking moments about A, calculate the weight W of the beam.

$W =$ N [3]

- (c) The metal of the wire in (b) has a Young modulus of 2.0×10^{11} Pa. Initially the bucket is empty. When the bucket is filled with paint of weight 78 N, the strain of the wire increases by 7.5×10^{-4} . The wire obeys Hooke's law.

Calculate, for the wire,

- (i) the increase in stress due to the addition of the paint,

increase in stress = Pa [2]

- (ii) its diameter.

diameter = m [3]

(a) Fig. 2.1 shows a liquid in a cylindrical container.

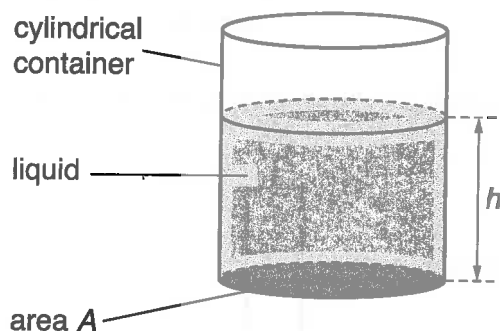


Fig. 2.1

The cross-sectional area of the container is A . The height of the column of liquid is h and the density of the liquid is ρ .

Show that the pressure p due to the liquid on the base of the cylinder is given by

$$p = \rho gh.$$

(b) The variation with height h of the total pressure P on the base of the cylinder in (a) is shown in Fig. 2.2. [3]

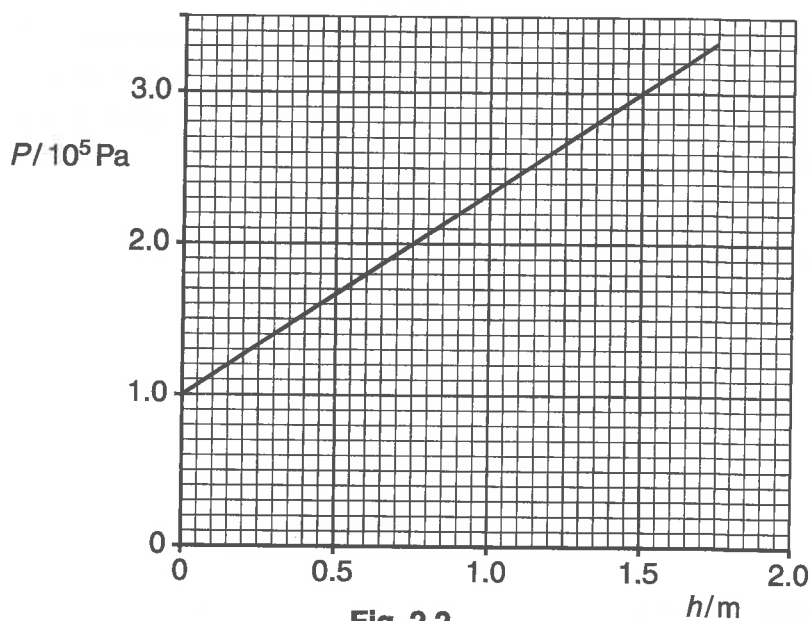


Fig. 2.2

(i) Explain why the line of the graph in Fig. 2.2 does not pass through the origin (0,0).

..... [1]

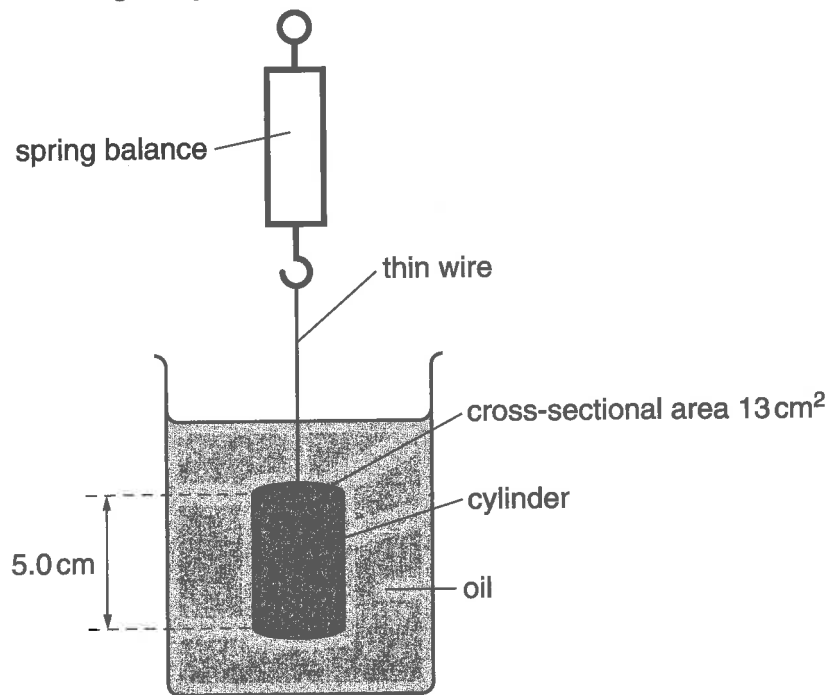
(ii) Use data from Fig. 2.2 to calculate the density of the liquid in the cylinder.

density = kg m⁻³ [2]

56. M/J 16/P23/Q4

booklet (scan + OCR)

A spring balance is used to weigh a cylinder that is immersed in oil, as shown in Fig. 4.1.

**Fig. 4.1**

The reading on the spring balance is 4.8 N. The length of the cylinder is 5.0 cm and the cross-sectional area of the cylinder is 13 cm². The weight of the cylinder is 5.3 N.

- (a) The cylinder is in equilibrium when it is immersed in the oil. Explain this in terms of the forces acting on the cylinder.

..... [1]

- (b) Calculate the density of the oil.

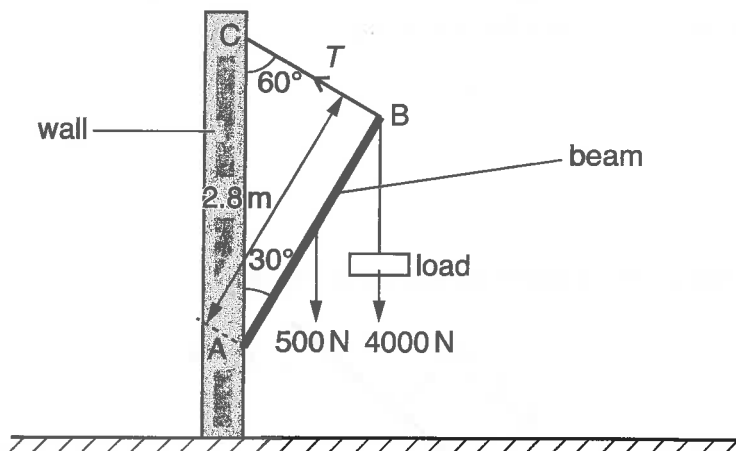
density = kg m⁻³ [3]

57. O/N 15/P22/Q4

booklet (scan + OCR)

(a) Define *moment of a force*.

[1]

(b) An arrangement for lifting heavy loads is shown in Fig. 4.1.**Fig. 4.1**

A uniform metal beam AB is pivoted on a vertical wall at A. The beam is supported by a wire joining end B to the wall at C. The beam makes an angle of 30° with the wall and the wire makes an angle of 60° with the wall.

The beam has length 2.8 m and weight of 500 N . A load of 4000 N is supported from B. The tension in the wire is T . The beam is in equilibrium.

(i) By taking moments about A, show that T is 2.1 kN .

[2]

(ii) Calculate the vertical component T_v of the tension T .

$$T_v = \dots\dots\dots \text{ N [1]}$$

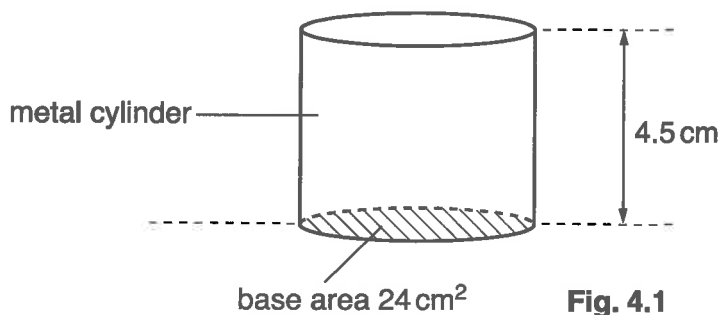
(iii) State and explain why T_v does not equal the sum of the load and the weight of the beam although the beam is in equilibrium.

.....
 [2]

58. O/N 15/P21/Q4

booklet (scan + OCR)

Fig. 4.1 shows a metal cylinder of height 4.5 cm and base area 24 cm^2 .

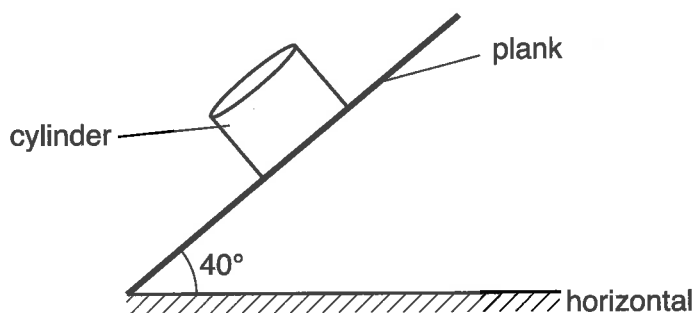
**Fig. 4.1**

The density of the metal is 7900 kg m^{-3} .

(a) Show that the mass of the cylinder is 0.85 kg.

[2]

(b) The cylinder is placed on a plank, as shown in Fig. 4.2.

**Fig. 4.2**

The plank is at an angle of 40° to the horizontal.

Calculate the pressure on the plank due to the cylinder.

pressure = Pa [3]

(c) The cylinder then slides down the plank with a constant acceleration of 3.8 ms^{-2} .

A constant frictional force f acts on the cylinder.

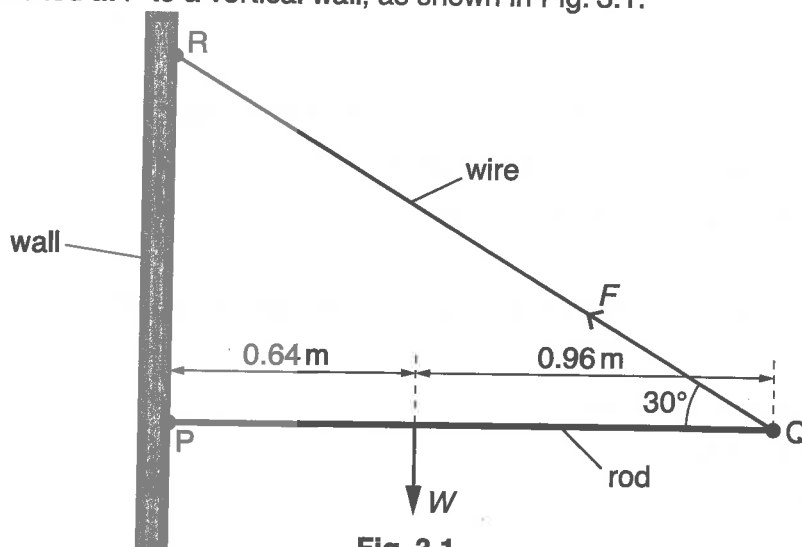
Calculate the frictional force f .

$f = \dots\dots\dots$ N [3]

59. M/J 15/P22/Q3

booklet (scan + OCR)

A rod PQ is attached at P to a vertical wall, as shown in Fig. 3.1.

**Fig. 3.1**

The length of the rod is 1.60 m. The weight W of the rod acts 0.64 m from P. The rod is kept horizontal and in equilibrium by a wire attached to Q and to the wall at R. The wire provides a force F on the rod of 44 N at 30° to the horizontal.

(a) Determine

(i) the vertical component of F ,

vertical component = N [1]

(ii) the horizontal component of F .

horizontal component = N [1]

(b) By taking moments about P, determine the weight W of the rod.

$W =$ N [2]

(c) Explain why the wall must exert a force on the rod at P.

..... [1]

(d) On Fig. 3.1, draw an arrow to represent the force acting on the rod at P. Label your arrow with the letter S. [1]

Answers Section

1. M/J 26/P22/Q2/a,b

2(a)	resultant force (in any direction) is zero	B1	Allow 'net force is zero' Allow 'sum of forces is zero' or 'total force is zero' or ' $\Sigma F = 0$ ' Allow 'sum of vertical forces is zero <u>and</u> sum of horizontal forces is zero'. Allow 'resultant force about any point is zero' Ignore 'forces balance' or 'forces cancel' or 'forces are equal' Ignore just 'no (external) forces act' or just 'force is zero' Ignore object stationary / constant velocity / zero acceleration etc
	resultant torque / moment (about any point) is zero	B1	Allow 'net moment is zero' Allow 'sum of moments is zero' or 'total moment is zero' Allow 'sum of CW moments = sum of ACW moments' Allow 'sum of CW and ACW moments is zero' Allow 'resultant moment at a point is zero' Allow 'torque' instead of 'moment' throughout Ignore 'moments balance' or 'moments cancel' Ignore 'no moments' or 'moment is zero' Not 'momentum' instead of 'moment'
2(b)(i)	one equation for forces on either block or plank: $2T\cos\theta = R + W_P$ or $2T\cos\theta + R = W_B$	M1	A full equation for the forces on either object must be seen Allow $0 = R + W_P - 2T\cos\theta$ Allow $0 = W_B - R - 2T\cos\theta$ Ignore just $R + W_P - 2T\cos\theta = W_B - R - 2T\cos\theta$ Allow $\sin(90 - \theta)$ for $\cos\theta$ Not $\sin\theta$ for $\cos\theta$
	both equations for forces on block and plank and ($2T\cos\theta =$) $R + W_P = W_B - R$	A1	Second mark for both separate equations seen and 'show that' answer given. <u>Special case:</u> $2T\sin\theta = R + W_P$ <u>and</u> $2T\sin\theta + R = W_B$ <u>and</u> ($2T\sin\theta =$) $R + W_P = W_B - R$ credit 1 / 2
2(b)(ii)	$2R = 8.1 \times 9.81 - 4.5 \times 9.81$	C1	Allow $2R = 8.1g - 4.5g$
	$R = 18 \text{ N}$	A1	Allow $g = 10$

2. M/J 26/P23/Q2/b(i)

2(b)(i)	$\rho = m / V$ $V = 0.018 / 960 = 1.875 \times 10^{-5}$	C1	Allow d or D for density
	(length =) $0.018 / (960 \times 2.6 \times 10^{-6}) = 7.2 \text{ (m)}$	A1	Full substitution and answer needed.

3. M/J 26/P24/Q4/c

4(c)	$W = 0.070 \times 9.81$ $= 0.69$	C1	treat use of $g = 10$ as TE unless already noted (2(b)(ii)) ecf for second and third marks (gives 0.59 N)
	$U = 1100 \times 9.81 \times 1.0 \times 10^{-5}$ $= 0.11$	C1	C1 marks are independent
	$D = 0.69 - 0.11$ $= 0.58 \text{ N}$	A1	No ecf for incorrect expression in 4(b)

4. MAR 26/P22/Q2/c

2(c)(i)	(upthrust =) $(160 + 23) \times 9.81 = 1800 \text{ (N)}$	A1
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2(c)(ii)	$\rho = F/gV$	C1
	$= 1800 / (9.81 \times 1.80 \times 10^5 \times 10^{-6})$	
	$= 1.02 \times 10^3 \text{ kg m}^{-3}$	A1
	OR $\rho = m/V$	(C1)
	$= 183 / (1.80 \times 10^5 \times 10^{-6})$	
	$= 1.02 \times 10^3 \text{ kg m}^{-3}$	(A1)
2(b)(i)	23×3.4 or $(23 + 160)v$	C1
	$23 \times 3.4 + (160 \times 0) = (23 + 160)v$	
	$v = 0.43 \text{ ms}^{-1}$	A1
2(b)(ii)	$E_{(K)} = \frac{1}{2} mv^2$	C1
	$E_2/E_1 = \frac{1}{2} (160+23) \times 0.43^2 / \frac{1}{2} (23) \times 3.4^2$ $= 16.9 / 132.9$	C1
	$E_2/E_1 = 0.13$	A1
2(b)(iii)	The <u>total</u> kinetic energy after the collision is less than the <u>total</u> kinetic energy before the collision, so inelastic	B1
	OR	(B1)
	The relative speed of approach is more than the relative speed of separation, so inelastic	
	OR $E_2 < E_1$ so inelastic	(B1)

5. MAR 26/P22/Q4/a(i,ii,iii)

4(a)(i)	product of force and <u>perpendicular</u> distance (between the line of action of the force and the point)	B1
4(a)(ii)	(the line of action of) the <u>weight</u> passes through P / the perpendicular distance of (the line of action of) the <u>weight</u> from P is zero (so the moment is zero)	B1
4(a)(iii)	$2T \sin \theta = W$	C1
	$\sin \theta = 19 / (2 \times 13)$	A1
	$\theta = 47^\circ$	

6. O/N 25/P21/Q2/a,c(i)

2(a)(i)	(normal) force per (unit cross-sectional) area	B1
2(a)(ii)	(due to difference in depth there is a) difference in pressure between top and bottom (of ball)	B1
	(due to pressure difference, upwards) force on bottom of ball is greater (than downwards force on top of ball, so resultant force is upwards)	B1
2(c)(i)	upthrust $= 920 \times (4/3) \times \pi \times (4.2 \times 10^{-3})^3 \times 9.81 = 2.8 \times 10^{-3} \text{ (N)}$	A1

7. O/N 25/P22/Q1/b(i,ii)

1(b)(i)	upthrust $= 1.1 \times 9.81 \times (4\pi \times 0.90^3 / 3)$	C1
	$= 33 \text{ N}$	A1
1(b)(ii)	(due to difference in height / depth there is a) difference in pressure between top and bottom (of balloon)	B1
	(due to pressure difference, upwards) force on bottom of balloon is greater (than downwards force on top of balloon, so resultant force is upwards)	B1

8. O/N 25/P22/Q2/a

2(a)(i)	$60 \times 1.6 (= 96 \text{ N m})$ or $220 \times 0.40 (= 88 \text{ N m})$	C1
	resultant moment = $96 - 88$ = 8.0 N m	A1
2(a)(ii)	the resultant force (of A and C) is not zero or forces (from A and C) are not parallel or forces (from A and C) are not equal (magnitude) or forces (from A and C) are not opposite (direction) or forces (from A and C) act through the same point so (they are) not a couple	B1

9. O/N 25/P22/Q3/a,b(i)

3(a)	an arrow horizontally on the page to the left labelled F	B1
	two arrows horizontally on the page to the right each labelled T	B1
3(b)(i)	$T = 110 / 2$ = 55 N	A1

10. O/N 25/P24/Q2/a,b,c,d,e

2(a)	product of force and distance	M1
	perpendicular distance between the (line of action of the two) forces	A1
2(b)	straight vertical line drawn from centre to bottom corner	B1
2(c)(i)	centre of gravity is to the right of the rod so torque is anticlockwise or moment of weight of sheet about rod is clockwise so torque is anticlockwise	B1
2(c)(ii)	(horizontal displacement) = $3.3 / (2.8 \times 9.81) = 0.12 \text{ (m)}$	A1
2(d)	$\rho = m / V$	C1
	$V = 4.0 \times 10^{-3} \times (\text{side length})^2$	C1
	side length = $\sqrt{2.8 / (3000 \times 0.0040)} = 0.48 \text{ m}$	A1
2(e)	point lies on a straight line at 45° to the horizontal from the bottom-right corner to the edge of the rod	B1
	point lies on a vertical line half-way between rod and right-hand edge	B1

11. M/J 25/P21/Q2/a,b

2(a)	in (rotational) equilibrium	B1
	<u>sum / total</u> of clockwise moments about a point = <u>sum / total</u> of anticlockwise moments about the (same) point.	B1
2(b)	$80 \times 9.81 \times 3$ or $60 \times 9.81 \times 3$ or $45 \times 9.81 \times x$	C1
	$80 \times 9.81 \times 3 = (60 \times 9.81 \times 3) + (45 \times 9.81 \times x)$	C1
	$x = 1.3 \text{ m}$	A1

12. M/J 25/P22/Q2/a,b

2(a)	force $\times$ <u>perpendicular</u> distance (of line of action of force to / from the point)	B1
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2(b)(i)	(moment due to weight =) $1.2 \times 270 \times 9.81$	B1
	(moment due to post =) $1800 \times (1.6 / \cos \theta)$	B1
	$1.2 \times 270 \times 9.81 = 1800 \times (1.6 / \cos \theta)$ so $\theta = 25(^{\circ})$ or $1.2 \times 270 \times 9.81 - 1800 \times (1.6 / \cos \theta) = 0$ so $\theta = 25(^{\circ})$	B1
2(b)(ii)	A closed tip-to-tail vector triangle	M1
	vector labelled F at an angle of $25^{\circ} \pm 3^{\circ}$ anticlockwise from vertical and vector labelled R at an angle of $37^{\circ} \pm 3^{\circ}$ clockwise from vertical	A1
2(b)(iii)	$A = F / p$ $= 1800 / (150 \times 10^3)$	C1
	$= 0.012 \text{ m}^2$	A1

13. M/J 25/P23/Q2/a,b(i)

2(a)	force $\times$ <u>perpendicular</u> distance (of line of action of force to / from the point)	B1
2(b)(i)	$150 \times 9.81 \times 4.5$ or $90 \times 9.81 \times 4.5$ or $3.0 \times 9.81 \times m$	C1
	$(150 \times 9.81 \times 4.5) = (90 \times 9.81 \times 4.5) + (3.0 \times 9.81 \times m)$	C1
	$m = 90 \text{ kg}$	A1

14. M/J 25/P24/Q1/a,b(i,iii)

1(a)	force $\times$ <u>perpendicular</u> distance (of line of action of force to a point) or product of force and perpendicular distance (to a point)	B1
1(b)(i)	in (rotational) equilibrium	B1
	<u>sum / total</u> of CW moments about a point = <u>sum / total</u> of ACW moments about the (same) point.	B1
1(b)(iii)	$F \times 80$ or 56×40 or $F \times 0.8$ or 56×0.4 $F \times 80 = 56 \times 40$	C1
	$F = 28 \text{ N}$	A1

15. MAR 25/P22/Q1/b(i)

1(b)(i)	$(\rho =) m / V$	C1
	$= 31.3 \times 10^{-3} / (1.53 \times 10^{-2})^3 = 8.7 \times 10^3 \text{ (kg m}^{-3}\text{)}$	A1

16. MAR 25/P22/Q2/a,b

2(a)	in (rotational) equilibrium	B1
	<u>sum / total</u> of CW moments about a point = <u>sum / total</u> of ACW moments about the (same) point.	B1
2(b)(i)	The magnitudes of the moments about A are: (1700×3.0) $(660 \times x)$ (1300×5) Correct magnitude of any one moment about A.	C1
	Correct magnitudes of a second moment about A.	C1
	$(1700 \times 3.0) + (660 \times x) = (1300 \times 5)$	A1
	$x = 2.1 \text{ m}$	
2(b)(ii)	(Upthrust =) $11 \times 9.81 + 1300 = 1400 \text{ (N)}$	A1

2(b)(iii)	$y = \text{Upthrust} / \rho g A$	C1
	$= 1400 / (990 \times 9.81 \times \pi \times (0.78 / 2)^2)$	
	$= 0.30 \text{ m}$	A1
	OR $[P = F / A = 1400 / \pi \times (0.78 / 2)^2 = 2930]$ $y = P / \rho g$ $= 2930 / (990 \times 9.81)$	(C1)
	$= 0.30 \text{ m}$	(A1)
2(b)(iv)	line with a non-zero value of depth at distance = 0	B1
	A line from distance = 0 to distance = 6.0 m with a gradient that is always positive	B1

17. O/N 24/P21/Q1/a,b(i)

1(a)	mass per unit volume	B1
1(b)(i)	density $= 0.243 / (0.0541 \times 0.1109 \times 0.0162)$	C1
	$= 2500 \text{ kg m}^{-3}$	A1

18. O/N 24/P22/Q3/b,c(i)

3(b)	resultant force (in any direction) is zero	B1
	resultant moment / torque (about any point) is zero	B1
3(c)(i)	(moment =) $33 \times 0.65 / 2$ or $1.5 \times (0.65 - 0.12)$ or $T \sin 50^\circ \times (0.65 / 2)$	C1
	sum of clockwise moments = sum of anticlockwise moments $33 \times (0.65 / 2) + 1.5 \times (0.65 - 0.12) = T \sin 50^\circ \times (0.65 / 2)$	C1
	tension = 46 N	A1

19. O/N 24/P23/Q3/a,b,c

3(a)	(for a system in equilibrium,) sum of clockwise moments (about a point) equals sum of anticlockwise moments (about the same point)	B1
3(b)(i)	2.6×0.40 or $F \times 0.40$ (any one moment)	C1
	$U = 4.0 - F$	A1
	$= 4.0 - (2.6 \times 0.40 / 0.40) = 1.4 \text{ N}$	
	or 2.6×0.4 or $(4.0 - U) \times 0.4$	(C1)
	$2.6 \times 0.4 = (4.0 - U) \times 0.4$ hence $U = (4.0 - 2.6) = 1.4 \text{ N}$	(A1)
3(b)(ii)	$U = \rho g V$ and $A = V / h$ or $p = h \rho g$ and $A = U / p$ or $A = U / h \rho g$	C1
	$A = 1.4 / (0.10 \times 1.0 \times 10^3 \times 9.81)$	C1
	$= 1.4 \times 10^{-3} \text{ m}^2$	A1

3(c)	line starting at (0.10, 0.40)	B1
	straight line with negative gradient	B1
	line ending at (0.29, 0)	B1

20. M/J 24/P22/Q4/c

4(c)(i)	$1.7 \times 2.0 (\times 10^{-2})$ or $4.5 \times 10^{-2} \times 9.81 \times d$	C1
	$1.7 \times 2.0 \times 10^{-2} = 4.5 \times 10^{-2} \times 9.81 \times d$	A1
	$d = 7.7 \times 10^{-2} \text{ m}$	
4(c)(ii)	clockwise	B1

21. M/J 24/P23/Q1/d(i)

1(d)(i)	$V = (4/3) \pi r^3$	C1
	upthrust = $(4/3) \times \pi \times 0.03^3 \times 920 \times 9.81 = 1.0 \text{ N}$	A1

22. MAR 24/P22/Q2/c(ii)

2(c)(ii)	$(F=) \rho g V$ $= 1000 \times 9.81 \times 7.5 \times 10^{-2} = 740 \text{ (N)}$	A1
2(b)	$\frac{1}{2} m(\Delta) v^2 = mg(\Delta) h$	C1
	$v^2 = 5.9^2 + 2 \times 9.81 \times 7.8$ $v^2 = 188$	C1
	$v = 14 \text{ m s}^{-1}$	A1
	or by resolving components Vertically: $v^2 = u^2 + 2as$ $v^2 = (5.9 \sin 60)^2 + 2 \times -9.81 \times (1.2 - 9.0)$ $v_v = 13.4$	(C1)
	horizontally: $v_h = 5.9 \cos 60$ $v_h = 2.95$	(C1)
	resultant velocity = $\sqrt{(13.4^2 + 2.95^2)}$ $= 14 \text{ m s}^{-1}$	(A1)

23. O/N 23/P21/Q2/a(i,ii)

2(a)(i)	arrow upwards ($\uparrow$) and labelled upthrust / U arrow downwards ($\downarrow$) and labelled weight / W / mg arrow downwards ($\downarrow$) and labelled tension / T <i>1 mark: One or two correctly labelled arrows</i> <i>2 marks: Three correctly labelled arrows</i>	B2
2(a)(ii)	$U = T + W$ or upthrust = tension + weight	C1
	$\rho V g = T + W$	C1
	$V = [(4.00 \times 10^2) + (3.39 \times 10^4)] / (1.23 \times 9.81)$	
	$V = 2.84 \times 10^3 \text{ m}^3$	A1

24. O/N 23/P22/Q2/a

2(a)	$65 = \rho g V$ or $65 = mg$ and $m = \rho V$	C1
	$\rho = 65 / (9.81 \times 7.5)$ $= 0.88 \text{ kg m}^{-3}$	A1

25. O/N 23/P23/Q1/b(ii)

1(b)(ii)	$\rho = (4 \times 0.247) / [\pi \times (26.2 \times 10^{-3})^2 \times 0.162]$	C1
	$\rho = 2.83 \times 10^3 \text{ kg m}^{-3}$	A1

26. O/N 23/P23/Q3/a,b

3(a)	resultant force (in any direction) is zero	B1
	resultant moment / torque (about any point) is zero	B1
3(b)(i)	$F = \rho V g$ $V = 93\,000 / (1.2 \times 9.81)$ $= 7900 \text{ m}^3$	A1
3(b)(ii)	weight = $93\,000 + 3(.0) \times 10^3$	C1
	$m = (93\,000 + 3.0 \times 10^3) / 9.81$ $= 9800 \text{ kg}$	A1

27. M/J 23/P22/Q2

- (a) the point where (all) the weight (of the object) is taken to act [1]
- (b) (i) (54×0.45) or (2.4×0.95) or $(T \sin 30^\circ \times 1.3)$
 $(54 \times 0.45) = (2.4 \times 0.95) + (T \sin 30^\circ \times 1.3)$
 $T = 34 \text{ N}$ [3]
- (ii) resultant moment $= (54 \times 0.45) - (2.4 \times 0.95)$ or $(34 \sin 30^\circ \times 1.3)$
 $= 22 \text{ N m}$ [1]
- (c) (i) $(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$
 $= 2.4 \times 1.8$
 $= 4.3 \text{ J}$ [2]
- (ii) $E = \frac{1}{2}mv^2$
 $= \frac{1}{2} \times (2.4 / 9.81) \times 3.4^2$
 $= 1.4 \text{ J (at X)}$
kinetic energy at Y $= 4.3 + 1.4$
 $= 5.7 \text{ J}$
- or
- $\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mg(\Delta)h$
 $v^2 = 3.4^2 + 2 \times 9.81 \times 1.8$
 $v^2 = 46.9$ so $v = 6.85 \text{ (ms}^{-1}\text{)}$
 $\text{KE} = \frac{1}{2} \times (2.4 / 9.81) \times 6.85^2$
 $= 5.7 \text{ J}$ [3]
- (iii) no variation or acceleration is (always) vertically downwards [1]
- (iv) horizontal straight line at a non-zero value of velocity [1]

28. M/J 23/P21/Q4

- (a) $\rho = m / V$ or $\rho = m / Ah$
 $p = F / A$ or $p = W / A$
appropriate algebra leading to $p = \rho gh$
e.g. $p = \rho Ahg / A$ or $\rho Vg / A$ or $\rho Vg / (V/h)$ and (so) $p = \rho gh$ [3]
- (b) there is atmospheric / air pressure [1]
- (c) $\Delta p = \rho g \Delta h$
e.g. $(9.66 - 9.60) \times 10^4 / 8.0 \times 10^{-2} = \rho \times 9.81$
 $\rho = 760 - 770 \text{ kg m}^{-3}$ [2]
- (d) $F = \rho g V$
 $= 760 \times 9.81 \times 3.7 \times 10^{-4} \times 4.0 \times 10^{-2} (= 0.11 \text{ N})$
tension $= 0.53 - 0.11$
 $= 0.42 \text{ N}$
- or
- $F = (\Delta)p \times A$
 $= (9.63 - 9.60) \times 10^4 \times 3.7 \times 10^{-4} (= 0.11 \text{ N})$
tension $= 0.53 - 0.11$
 $= 0.42 \text{ N}$ [2]

29. M/J 23/P23/Q2

(a) $F = 1050 \times 9.81 \times 0.21V$ or $W = \rho \times 9.81 \times V$

$$0.21V \times 1050 (\times 9.81) = V (\times 9.81) \times \rho$$

$$\rho = 220 \text{ kg m}^{-3}$$

[2]

(b) (i) $F = 1050 \times 9.81 \times V$ or $W = 220 \times 9.81 \times V$

$$(V \times 1050 \times 9.81) - (V \times 220 \times 9.81) = (V \times 220) \times a$$

$$a = 37 \text{ ms}^{-2}$$

[3]

- (ii) the (downward) drag / viscous force increases (with speed)
resultant force decreases (as upthrust and weight remain the same)
acceleration decreases (as its velocity increases)

[3]

30. O/N 22/P22/Q2

(a) $F = \rho g V$

$$0.071 = 1.2 \times 9.81 \times V$$

$$(V = 6.03 \times 10^{-3} \text{ m}^3)$$

$$\frac{4}{3} \times \pi \times r^3 = 6.03 \times 10^{-3}$$

$$r = 0.11 \text{ m}$$

[2]

(b) $m = 0.053 / 9.81$

$$(= 5.4 \times 10^{-3} \text{ kg})$$

$$F = 0.071 - 0.053$$

$$(= 0.018 \text{ N})$$

$$a = (0.071 - 0.053) / (0.053 / 9.81)$$

$$= 3.3 \text{ m s}^{-2}$$

[3]

(c) (i) $v^2 = u^2 + 2as$

$$3.6^2 = (-1.4)^2 + 2 \times 9.81 \times s \text{ or } 3.6^2 = 1.4^2 + 2 \times 9.81 \times s$$

$$s = 0.56 \text{ m}$$

[2]

- (ii) single straight line from any positive non-zero value of v at $t = 0$ to any
negative non-zero value of v at $t = T$ line starting at $(0, 1.4)$ and ending at $(T, -3.6)$

[2]

31. O/N 22/P22/Q3

(a) point where (all) the weight (of an object) is taken to act

[1]

(b) (i) vertical component $= 45 \sin 37^\circ$

$$= 27 \text{ N}$$

[1]

- (ii) the magnitudes of the three moments about A are (23×0.48) , (27×0.56) and $(W \times 0.76)$
 correct magnitude of any one moment about A
 correct magnitudes of any two moments about A
 $(23 \times 0.48) + (W \times 0.76) = 27 \times 0.56$
 $W = 5.4 \text{ N}$ [3]
- (iii) horizontal component $= 45 \cos 37^\circ$
 $= 36 \text{ N}$ [1]
- (iv) decrease [1]
- (v) $\sigma = F / A$
 $\sigma = F / \pi r^2$ or $4F / \pi d^2$
 so $\sigma = 5.3 \times 10^7 \times \pi r^2 / \pi (3r)^2$
 $= 5.3 \times 10^7 / 9$
 $= 5.9 \times 10^6 \text{ Pa}$ [2]

32. O/N 22/P23/Q3

- (a) sum of CW moments = sum of ACW moments
 about the same point for (an object in rotational) equilibrium [2]
- (b) moment $= 0.3(0) \times 0.29 \cos 40^\circ$ or $0.3(0) \times 0.222$
 $= 0.067 \text{ Nm}$ [2]
- (c) (i) $k = F / x$ or $k = \text{gradient}$
 e.g. $k = 21 / 10 \times 10^{-3}$
 $k = 2100 \text{ N m}^{-1}$ [2]
- (ii) $V_{\text{(sphere)}} = \frac{4}{3} \times \pi \times (0.0480)^3$
 $F = \rho g V$
 (upthrust $=$) $1000 \times 9.81 \times (\frac{4}{3} \times \pi \times (0.048)^3) \times 0.26(0) = 1.18 \text{ (N)}$ [2]
- (iii) 1.18×0.29 or 0.30×0.29 or $F \times 0.017$
 $(1.18 \times 0.29) = (0.30 \times 0.29) + (F \times 0.017)$
 $F = 15 \text{ N}$ [2]

33. M/J 22/P22/Q2

- (a) $T \sin 68^\circ + 32 = 280$
 $T = 270 \text{ N}$ [2]
- (b) (i) $F = \rho g V$
 $V = 280 / (1.0 \times 10^3 \times 9.81)$
 $= 0.029 \text{ m}^3$ [1]
- (ii) $\rho = (32 / 9.81) / 0.029$
 $= 110 \text{ kg m}^{-3}$ [2]
- (c) $(\Delta)E = mg(\Delta)h$ or $(\Delta)E = W(\Delta)h$
 $(\Delta)h = (-) 77 / 32$
 $(\Delta)h = (-) 2.4$
 final height $= 6.2 - 2.4$
 $= 3.8 \text{ m}$ [3]

(d) (i) $T = kx$ where k is a constant

or

$T = (EA/L)x$ where A is (cross-sectional) area, E is Young modulus, L is (original/unstretched) length

[1]

(ii) $E = \frac{1}{2}kx^2$

or

$$E = \frac{1}{2}Fx \text{ and } F = kx$$

$$E = 0.65 \times 2^2$$

$$= 2.6 \text{ J}$$

or

$$E = \frac{1}{2}Fx$$

$$0.65 = \frac{1}{2} \times 270 \times x \text{ and so } x = 4.8 \times 10^{-3} \text{ m}$$

$$k = F/x = 270 / 4.8 \times 10^{-3}$$

$$= 5.6 \times 10^4$$

final $E = \frac{1}{2} \times 540 \times 9.6 \times 10^{-3}$

or

$$E = \frac{1}{2} \times (270 \times 2) \times (4.8 \times 10^{-3} \times 2)$$

or

$$E = \frac{1}{2} \times 5.6 \times 10^4 \times (9.6 \times 10^{-3})^2$$

$$= 2.6 \text{ J}$$

[2]

34. M/J 22/P21/Q2

(a) point where (all) the weight (of the object) is taken to act

[1]

(b) (i) moments about P are: $(F \times 1.10)$, (44.0×0.60) and (3.0×2.90)

1 mark for any one correct moment and 2 marks for two correct moments

$$(F \times 1.10) = (44.0 \times 0.60) + (3.0 \times 2.90)$$

$$F = 32 \text{ N}$$

[3]

(ii) force = $32 + 44 + 3$

$$= 79 \text{ N}$$

[1]

(c) (i) $(F = \rho g V)$

$$V = 2.5 / (1100 \times 9.81) (= 2.32 \times 10^{-4})$$

$$V = (4/3)\pi r^3$$

$$r = [(3 \times 2.32 \times 10^{-4}) / 4\pi]^{1/3}$$

$$r = 0.038 \text{ m}$$

[3]

(ii) resultant moment = 2.5×2.9

$$= 7.3 \text{ N m}$$

or

$$\text{resultant moment} = F \times 1.1 - (44 \times 0.6 + 0.5 \times 2.9)$$

$$= 7.3 \text{ N m (allow 7.2 N m or 7.4 N m from rounded values of } F)$$

direction: anticlockwise

[2]

35. MAR 22/P22/Q3

- (a) $(m =) \rho V$
 $= 1.0 \times 10^3 \times 1.5 \times 10^{-4} \times 5.0 \times 1.6 = 1.2 \text{ (kg)}$ [2]
- (b) (i) $(\Delta)p = 1.2 \times 5.0$
 $= 6.0 \text{ N s}$ [1]
- (ii) $F = 6.0 / 1.6$ or $= 1.2 \times 5.0 / 1.6$
 $= 3.8 \text{ N}$ [1]
- (c) Newton's third law applies (so) 3.8 N. [1]
- (d) $p = F / A$
 $= 3.8 / 1.5 \times 10^{-4}$
 $= 2.5 \times 10^4 \text{ Pa}$ [2]

36. O/N 21/P23/Q3

- (a) change in displacement / time (taken) [1]
- (b) by calculation:
 $v^2 = 42^2 + 23^2 - (2 \times 42 \times 23 \times \cos 54^\circ)$ or
 $v^2 = (42 - 23 \cos 54^\circ)^2 + (23 \sin 54^\circ)^2$ or
 $v^2 = (42 - 23 \sin 36^\circ)^2 + (23 \cos 36^\circ)^2$
 $v = 34 \text{ m s}^{-1}$ or
 by scale diagram: triangle of vector velocities drawn
 $v = 34 \text{ m s}^{-1}$ (allow $\pm 1 \text{ m s}^{-1}$ if scale diagram used) [2]
- (c) (i) $(\Delta)E = mg(\Delta)h$ or $(\Delta)E = W(\Delta)h$
 $h = 6100 / 46 (= 133 \text{ m})$
 $\theta = \sin^{-1} (133 / 280)$
 $= 28^\circ$ [3]
- (ii) force = $6100 / 280$ or $46 \sin 28^\circ$
 $= 22 \text{ N}$ [2]
- (d) $v_{(s)} = 280 / 14 (= 20 \text{ m s}^{-1})$
 $f_o = f_s v / (v - v_s)$
 $f_s = 450 \times (340 - 20) / 340$
 $= 420 \text{ Hz}$ [3]

37. M/J 21/P21/Q3

- (a) solid straight line drawn between centre of sphere at X and at Y [1]
- (b) $p = mv$ or $0.72 = mv$
 $E = \frac{1}{2}mv^2$ or $0.86 = \frac{1}{2}mv^2$
 $(m =) 0.72^2 / (2 \times 0.86) = 0.30 \text{ (kg)}$ or
 $v = 2E_K / p$
 $v = (0.86 \times 2) / 0.72 = 2.4 \text{ (to 2 s.f.)}$
 $m = 0.72 / 2.4 = 0.30 \text{ (kg)}$ [3]
- (c) $(\Delta)E = mg(\Delta)h$
 $h = 0.86 / (0.30 \times 9.81)$
 $= 0.29 \text{ m}$ [2]
- (d) $\cos \theta = (0.93 - 0.29) / 0.93$ so $\theta = 47^\circ$ [1]
- (e) moment = $(0.30 \times 9.81) \times 0.93 \times (\sin 47^\circ \text{ or } \cos 43^\circ)$ or

$$\begin{aligned}\text{moment} &= (0.30 \times 9.81) \times [0.93^2 - (0.93 - 0.29)^2]^{0.5} \\ &= 2.0 \text{ N m}\end{aligned}$$

- (f) there is a resultant force (acting on sphere) or
there is a resultant moment (about P acting on pendulum)
(so) not in equilibrium

[2]

[1]

38. M/J 21/P23/Q1

- (a) (i) two correct scalar quantities e.g. time, mass, distance, temperature
two correct vector quantities e.g. force, acceleration, velocity, displacement [2]
(ii) magnitude
unit [2]
(b) (i) north component of velocity = 11 m s^{-1}
east component of velocity = 7.5 m s^{-1} [2]
(ii) velocity = $7.5 - 2.7$
 $= 4.8 \text{ m s}^{-1}$ [1]
(iii) velocity = $\sqrt{(11^2 + 4.8^2)}$
 $= 12 \text{ m s}^{-1}$ [2]
(iv) angle = $\tan^{-1}(4.8 / 11)$
 $= 24^\circ$ [2]

[2]

39. O/N 20/P22/Q2

- (a) (i) $(\Delta)p = \rho g(\Delta)h$
 $520 = 1000 \times 9.81 \times h$
 $h = 0.053 \text{ m}$ [2]
(ii) (upthrust =) $(\Delta)p \times A$
 $= (\Delta)p \times \pi(d/2)^2$ or $(\Delta)p \times \pi r^2$
 $= 520 \times \pi(0.031/2)^2 = 0.39 \text{ (N)}$ [2]
(iii) $T = 0.84 - 0.39$
 $= 0.45 \text{ N}$ [1]
(b) (i) $a = (v - u) / t$ or $(\Delta)v / (\Delta)t$ or gradient
 $= \text{e.g. } 8.0 \times 10^{-2} / 2.0$
 $= 4.0 \times 10^{-2} \text{ m s}^{-2}$ [2]
(ii) distance = $(\frac{1}{2} \times 2.5 \times 0.10) + (\frac{1}{2} \times 1.5 \times 0.10)$ or $(\frac{1}{2} \times 4.0 \times 0.10)$
 $(= 0.20 \text{ (m)})$
depth = $0.32 - 0.20$
 $= 0.12 \text{ m}$ [2]
(c) (i) viscous (force) [1]
(ii) viscous force increases (with speed/time/depth) (so) acceleration decreases [2]

40. O/N 20/P22/Q4

- (a) (component =) $96 \sin 38^\circ = 59 \text{ (N)}$ or $96 \cos 52^\circ = 59 \text{ (N)}$ [1]
(b) (45×2.9) or $(T \times 1.8)$ or (59×1.5)
 $(45 \times 2.9) = (T \times 1.8) + (59 \times 1.5)$
 $T = 23 \text{ N}$ [3]

41. O/N 20/P21/Q1

- (a) (i) force $\times$ perpendicular distance (of line of action of force to the point) [1]
 (ii) units: kgms^{-2}m
 $= \text{kg m}^2 \text{s}^{-2}$ [1]
- (b) $W = \rho Vg$ or $W = \rho ALg$
 $A = 5.2 / (790 \times 2.4 \times 9.81)$
 $(= 2.8 \times 10^{-4} \text{ (m}^2\text{)})$
 $= 2.8 \times 10^2 \text{ mm}^2$ [3]
- (c) (i) (component $=$) $5.2 \sin 56^\circ = 4.3 \text{ (N)}$
 or
 $5.2 \cos 34^\circ = 4.3 \text{ (N)}$ [1]
- (ii) $(T \times 2.4)$ or (4.3×1.2) or (4.6×1.8)
 $(T \times 2.4) + (4.3 \times 1.2) = (4.6 \times 1.8)$
 $T = 1.3 \text{ N}$ [3]

42. O/N 20/P23/Q2

- (a) point where (all) the weight (of the body) is considered/seems to act [2]
 (b) (i) horizontal component of force $= 38 \cos 60^\circ$ or $38 \sin 30^\circ$
 $= 19 \text{ N}$ [1]
 (ii) $(T \times 1.2)$ or (19×0.9) or 17
 $(T \times 1.2) = (19 \times 0.9)$
 $T = 14 \text{ N}$ [2]
 (iii) $F = 45 + 38 \sin 60^\circ$
 $= 78 \text{ N}$ [1]

43. M/J 20/P23/Q3

- (a) for a body in (rotational) equilibrium
 sum/total of clockwise moments about a point = sum/total of anticlockwise moments about the (same) point [2]
- (b) (i) $(W \times 0.45)$ or (19×1.3) or $(W \times 1.85)$ or (22×2.6)
 $(W \times 0.45) + (19 \times 1.3) + (W \times 1.85) = (22 \times 2.6)$ so $W = 14 \text{ N}$ [2]
 (ii) magnitude $= 19 + 14 + 14 - 22$
 $= 25 \text{ N}$
 direction: vertically upwards [2]
- (c) (i) the extension is zero when the force is zero
 graph is a straight line and (so) Hooke's law obeyed [2]
 (ii) $k = F/x$ or $k = \text{gradient}$
 e.g. $k = 60 / (1.00 - 0.25)$
 $k = 80 \text{ N m}^{-1}$ [2]
 (iii) area shaded below graph line between $L = 0.25 \text{ m}$ and $L = 0.75 \text{ m}$ [1]

44. O/N 19/P22/Q4

(a) (i) (vertically) upwards/up [1]

(ii) increases (with time/velocity/depth) [1]

(b) (i) for a body in (rotational) equilibrium

sum/total of clockwise moments about a point = sum/total of anticlockwise moments about the (same) point [2]

(ii) ($F_B \times 5.0$) or (380×2.5) or (750×1.6)

$$(F_B \times 5.0) = (380 \times 2.5) + (750 \times 1.6)$$

$$F_B = 430 \text{ N}$$

[2]

(iii) taking moments about C:

$$(380 \times 2.5) = 750 \times (2.0 - x)$$

$$(2.0 - x) = 1.3$$

$$x = 0.7 \text{ m} \quad \text{or}$$

moments may be taken about other points, e.g. about D:

$$(380 \times 4.5) + (750 \times x) = 1130 \times 2.0$$

$$x = 0.7 \text{ m}$$

[2]

45. O/N 19/P23/Q1

(a) base units: $\text{kg m s}^{-2} \times \text{m}$
 $= \text{kg m}^2 \text{s}^{-2}$

[1]

(b) (i) distance of COG from P (= GP)

$$= 17 \cos 45^\circ - 4.0 \quad \text{or} \quad (144.5)^{1/2} - 4.0 \quad (= 8.0 \text{ cm})$$

$$\text{moment} = 0.15 \times 8.0 \times 10^{-2}$$

$$= 1.2 \times 10^{-2} \text{ Nm}$$

[2]

(ii) (line of action of) weight acts through pivot/P or

distance between (line of action of) weight and pivot/P is zero

(so) weight does not have a moment about pivot/P

[2]

46. M/J 19/P22/Q3

(a) the point where (all) the weight (of the body) is taken to act

[1]

(b) (i) vertical component = $54 \sin 35^\circ$

$$= 31 \text{ N}$$

[1]

(ii) the (line of action of the) force (at B) passes through (point) A or

the (line of action of the) force (at B) has zero (perpendicular) distance from (point) A [1]

(iii) $54 \sin 35^\circ \times 0.68$ or $54 \cos 35^\circ \times 0.68$ or $W \times 0.34$

$$54 \sin 35^\circ \times 0.68 + 54 \cos 35^\circ \times 0.68 = W \times 0.34 \quad \text{so} \quad W = 150 \text{ (N)}$$

[2]

(iv) total vertical force = $150 - 31$

$$= 120 \text{ N}$$

[1]

(c) $(\Delta)E = mg(\Delta)h$

$$E = \frac{1}{2}mv^2$$

$$\text{ratio} = (m \times 9.81 \times 4.8) / (\frac{1}{2} \times m \times 9.2^2) \quad \text{or} \quad (9.81 \times 4.8) / (\frac{1}{2} \times 9.2^2)$$

$$= 1.1$$

[4]

47. M/J 19/P23/Q3

- (a) $P = Fv$
 $P = 8.9 \cos 30^\circ \times 0.60$
 $= 4.6 \text{ W}$ [2]
- (b) $p = F/A$
 $F = 8.9 \sin 30^\circ + (0.24 \times 9.81)$
 $(= 6.80 \text{ N})$
 $A = 6.80 / 3500$
 $= 1.9 \times 10^{-3} \text{ m}^2$ [3]
- (c) (i) upwards/up [1]
(ii) the Earth/planet [1]

48. O/N 18/P22/Q2

- (a) ampere
kelvin
(allow mole, candela)
any two correct answers, 1 mark each [2]
- (b) (i) frictional (force)/friction [1]
(ii) $W \cos 31^\circ \times 3.0$ or 90×6.0
 $W \cos 31^\circ \times 3.0 = 90 \times 6.0$
 $W = 210 \text{ N}$ [2]
(iii) $X = 90 \sin 31^\circ$
 $= 46 \text{ N}$ [1]

49. M/J 18/P22/Q5

- (a) $\rho = m/V$
 $= (560/9.81)/(1.2 \times 0.018)$
 $= 2600 \text{ kg m}^{-3}$ [2]
- (b) $(\Delta)p = 940 \times 9.81 \times 1.2$
(upthrust $\Rightarrow$) $940 \times 9.81 \times 1.2 \times 0.018 = 200 \text{ N}$ [2]
- (c) (i) tension $= 560 - 200$
 $= 360 \text{ N}$ [1]
(ii) $P = Fv$
 $= 360 \times 0.020$
 $= 7.2 \text{ W}$ [2]
- (d) (i) upthrust decreases
tension (in wire) increases
power (output of motor) increases [3]
(ii) there is work done (on the cylinder) by the upthrust or GPE of oil decreases (as it fills the space left by cylinder and so total energy is conserved) [1]

50. M/J 18/P23/Q2

- (a) (work =) force $\times$ distance moved in the direction of the force [1]
- (b) (i) $\rho = m/V$
 $= (20/9.81)/(4/3 \times \pi \times 0.16^3)$
 $= 120 \text{ kg m}^{-3}$ [2]
- (ii) the pressure on the lower surface (of sphere) is greater than the pressure on the upper surface (of sphere) [1]
- (iii) $a = (170 - 20)/(20/9.81)$
 $= 74 \text{ ms}^{-2}$ [2]
- (iv) $D = 170 - 20 (= 150)$
 $810 \times (0.16^2) \times v^2 = 150$
 $v = 2.7 \text{ ms}^{-1}$ [3]
- (v) $4870 = (4850 \times v)/(v - 6.30)$
 $v = 1530 \text{ ms}^{-1}$ [2]

51. O/N 17/P22/Q2

- (a) force $\times$ perpendicular distance (of line of action of force) to/from a point [1]
- (b) (i) $2.4r$ or $(1.2 \times 2r)$ or $(1.2r + 1.2r)$ [1]
- (ii) (anticlockwise moment =) $6.0 \times r/2 \times \sin \theta$
 $6.0 \times r/2 \times \sin \theta = 2.4r$
 $\theta = 53^\circ$ [2]
- (iii) 6.0 N [1]

52. O/N 17/P23/Q2

- (a) $\rho = m/V$ or $\rho = m/Ah$
 $p = F/A$ or $p = W/A$
 $p = [\rho Ahg]/A$ or $p = [\rho Vg]/[V/h]$ (so) $p = \rho gh$ [3]
- (b) (i) 1. weight/gravitational (force)
upthrust (force)/buoyancy (force)
drag/viscous/frictional (force)/fluid resistance/resistance [2]
2. weight = upthrust + viscous (force)
- (ii) • decrease in (gravitational) potential energy (of sphere) due to decrease in height (since $E_p = mgh$)
• increase in thermal energy due to work done against viscous force/drag
- loss/change of (total) E_p equal to gain/change in thermal energy
Any 2 points. [2]
- (c) (i) atmospheric pressure $= 9.1(0) \times 10^4 \text{ Pa}$ [1]
- (ii) $(\Delta)p = \rho g(\Delta)h$
 $(9.15 - 9.10) \times 10^4 = \rho \times 9.81 \times (0.17 - 0.10)$
 $\rho = 730 \text{ (728) kg m}^{-3}$ [2]

53. M/J 17/P21/Q3

(a) $\rho = m/V$

$V = \pi d^2 L / 4 \text{ or } \pi r^2 L$

$\text{weight} = 2.7 \times 10^3 \times \pi (1.2 \times 10^{-2})^2 \times 5.0 \times 10^{-2} \times 9.81 = 0.60 \text{ N}$ [3]

(b) (i) the point from where (all) the weight (of a body) seems to act [1]

(ii) $W \times 12$

$(0.25 \times 8) + (0.6 \times 38)$

$W = (2 + 22.8) / 12$

$= 2.1 \text{ (2.07) N}$ [3]

(c) (i) pressure changes with depth (in water) or
pressure on bottom (of cylinder) different from pressure on top
pressure on bottom of cylinder greater than pressure on top or
force (up) on bottom of cylinder greater than force (down) on top [2](ii) anticlockwise moment reduced and reducing the weight of X reduces clockwise moment
or
anticlockwise moment reduced so clockwise moment now greater than (total)
anticlockwise moment [1]**54. O/N 16/P22/Q3**(a) resultant force (in any direction) is zero
resultant moment/torque (about any point) is zero [2]

(b) (i) force = $33 \sin 52^\circ$ or $33 \cos 38^\circ$
 $= 26 \text{ N}$ [1]

(ii) 26×0.30 or $W \times 0.20$ or 12×0.40

$26 \times 0.30 = (W \times 0.20) + (12 \times 0.40)$

$W = 15 \text{ N}$ [3]

(c) (i) $E = \Delta\sigma / \Delta\epsilon$ or $E = \sigma / \epsilon$

$\Delta\sigma = 2.0 \times 10^{11} \times 7.5 \times 10^{-4}$

$= 1.5 \times 10^8 \text{ Pa}$ [2]

(ii) $\Delta\sigma = \Delta F / A$ or $\sigma = F / A$

$A = 78 / 1.5 \times 10^8 (= 5.2 \times 10^{-7} \text{ m}^2)$

$5.2 \times 10^{-7} = \pi d^2 / 4$

$d = 8.1 \times 10^{-4} \text{ m}$ [3]

55. M/J 16/P22/Q2

(a) $p = F/A$

use of $m = \rho V$ and use of $V = Ah$ and use of $F = mg$ correct substitution to obtain $p = \rho gh$ [3](b) (i) (when h is zero the pressure is not zero due to) pressure from the
air/atmosphere [1]

(ii) gradient = ρg or $P - 1.0 \times 10^5 = \rho gh$

e.g. $\rho g = 1.0 \times 10^5 / 0.75 (= 133333)$

$\rho = 133333 / 9.81$

$= 14000 \text{ (13592) kg m}^{-3}$

[2]

56. M/J 16/P23/Q4

- (a) (resultant force = 0) (equilibrium)
therefore: weight – upthrust = force from thin wire (allow tension in wire) *or*
5.3 (N) – upthrust = 4.8 (N) [1]
- (b) difference in weight = upthrust *or* upthrust = 0.5 (N)
 $0.5 = \rho ghA$ *or* $m = 0.5/9.81$ and $V = 5.0 \times 13 \times 10^{-6} \text{ (m}^3\text{)}$
 $\rho = 0.5 / (9.81 \times 5.0 \times 13 \times 10^{-6})$
 $= 780 \text{ (784) kg m}^{-3}$ [3]

57. O/N 15/P22/Q4

- (a) product of the force and the perpendicular distance to/from a point/pivot [1]
- (b) (i) $4000 \times 2.8 \times \sin 30^\circ$ *or* $500 \times 1.4 \times \sin 30^\circ$ *or* $T \times 2.8$
or 4000×1.4 *or* 500×0.7
 $4000 \times 2.8 \times \sin 30^\circ + 500 \times 1.4 \times \sin 30^\circ = T \times 2.8$
hence $T = 2100 \text{ (2125) N}$ [2]
- (ii) ($T_v = 2100 \cos 60^\circ = 1100 \text{ (1050) N}$) [1]
- (iii) there is an upward (vertical component of) force at A
upward force at A + T_v = sum of downward forces/weight+load/4500 N [2]

58. O/N 15/P21/Q4

- (a) density = mass/volume
mass = $7900 \times 4.5 \times 24 \times 10^{-6} = 0.85 \text{ (0.853) kg}$ [2]
- (b) pressure = force/area
force = $W \cos 40^\circ$
pressure = $(0.85 \times 9.81 \cos 40^\circ) / 24 \times 10^{-4}$
 $= 2.7 \text{ (2.66) } \times 10^3 \text{ Pa}$ [3]
- (c) $F = ma$
 $W \sin 40^\circ - f = ma$
 $0.85 \times 9.81 \times \sin 40^\circ - f = 0.85 \times 3.8$
 $f (= 5.36 - 3.23) = 2.1 \text{ N}$ [5.38 – 3.242 if 0.8532 kg is used for the mass] [3]

59. M/J 15/P22/Q3

- (a) (i) (vertical component = $44 \sin 30^\circ = 22 \text{ N}$) [1]
(ii) (horizontal component = $44 \cos 30^\circ = 38(.1) \text{ N}$) [1]
- (b) $W \times 0.64 = 22 \times 1.60$
($W = 55 \text{ N}$) [2]
- (c) F has a horizontal component (not balanced by W)
or F has 38 N acting horizontally *or* 38 N acts on wall
or vertical component of F does not balance W
or F and W do not make a closed triangle of forces [1]
- (d) line from P in direction towards point on wire vertically above W and direction up [1]